Refreshing Your UI with HTML5, Bootstrap and CSS3

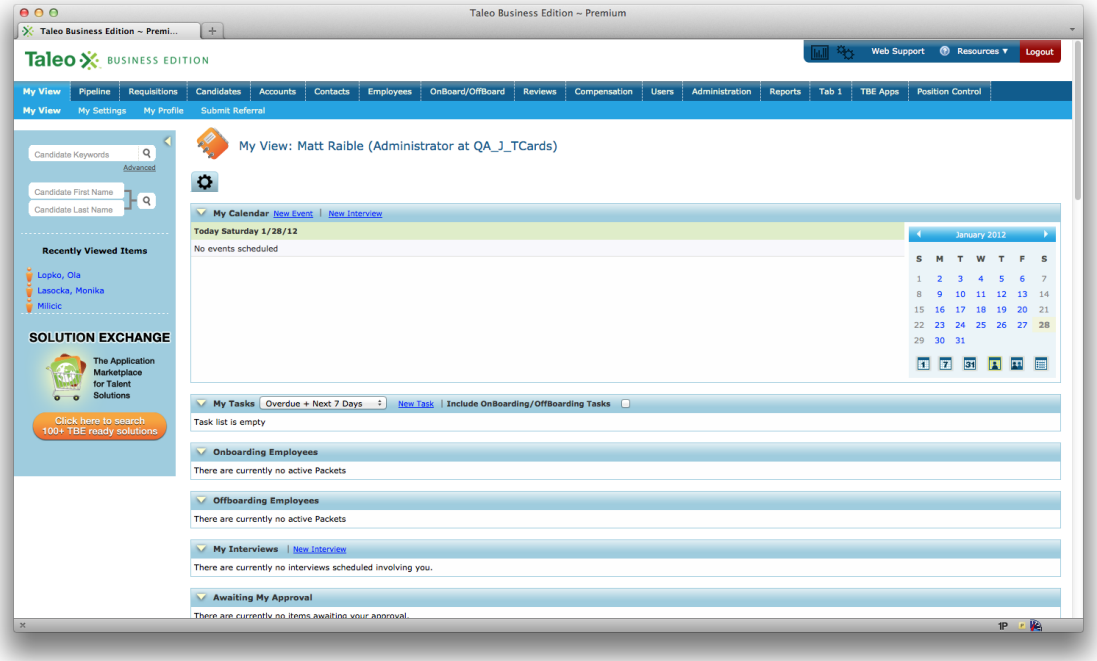
HTML5 Denver April 22, 2013

Matt Raible
http://raibledesigns.com

Photos by **Trish McGinity**

About Me

- Web Architecture Consultant
- Java Community Professional
- Father, Speaker, Beer Lover
- What Keeps me busy
  - Skiing, Mountain Biking
  - The Bus
  - AppFuse
  - Raible Designs
- **Contact Information**
  - www: http://raibledesigns.com
  - email: matt@raibledesigns.com
  - twitter: @mraible



Legacy Design

Have you worked with an outdated design?

My Experience

Taleo

- December 2011
- Page Speed Project
- UI Refresh Project
- Bootstrap
- Legacy Browsers

Best Practices for Speeding Up Your Web Site

Minimize HTTP Requests
Use a Content Delivery Network
Add an Expires or a Cache-Control Header
Gzip Components
Put Stylesheets at the Top
Put Scripts at the Bottom
Avoid CSS Expressions
Make JavaScript and CSS External
Reduce DNS Lookups
Minify JavaScript and CSS
No 404s

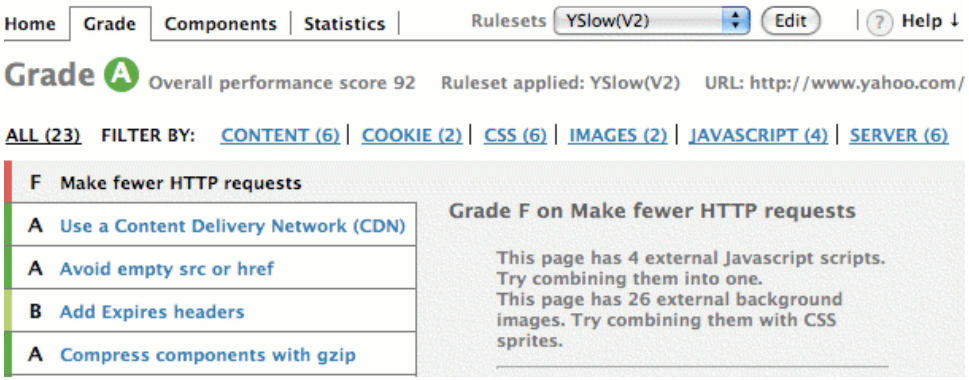
Avoid Redirects
Remove Duplicate Scripts
Configure ETags
Make Ajax Cacheable
Flush the Buffer Early
Use GET for AJAX Requests
Post-load Components
Split Components Across Domains
Optimize Images
Choose <link> over @import
Minimize the Number of iframes

35 Total Rules → http://developer.yahoo.com/performance/rules.html

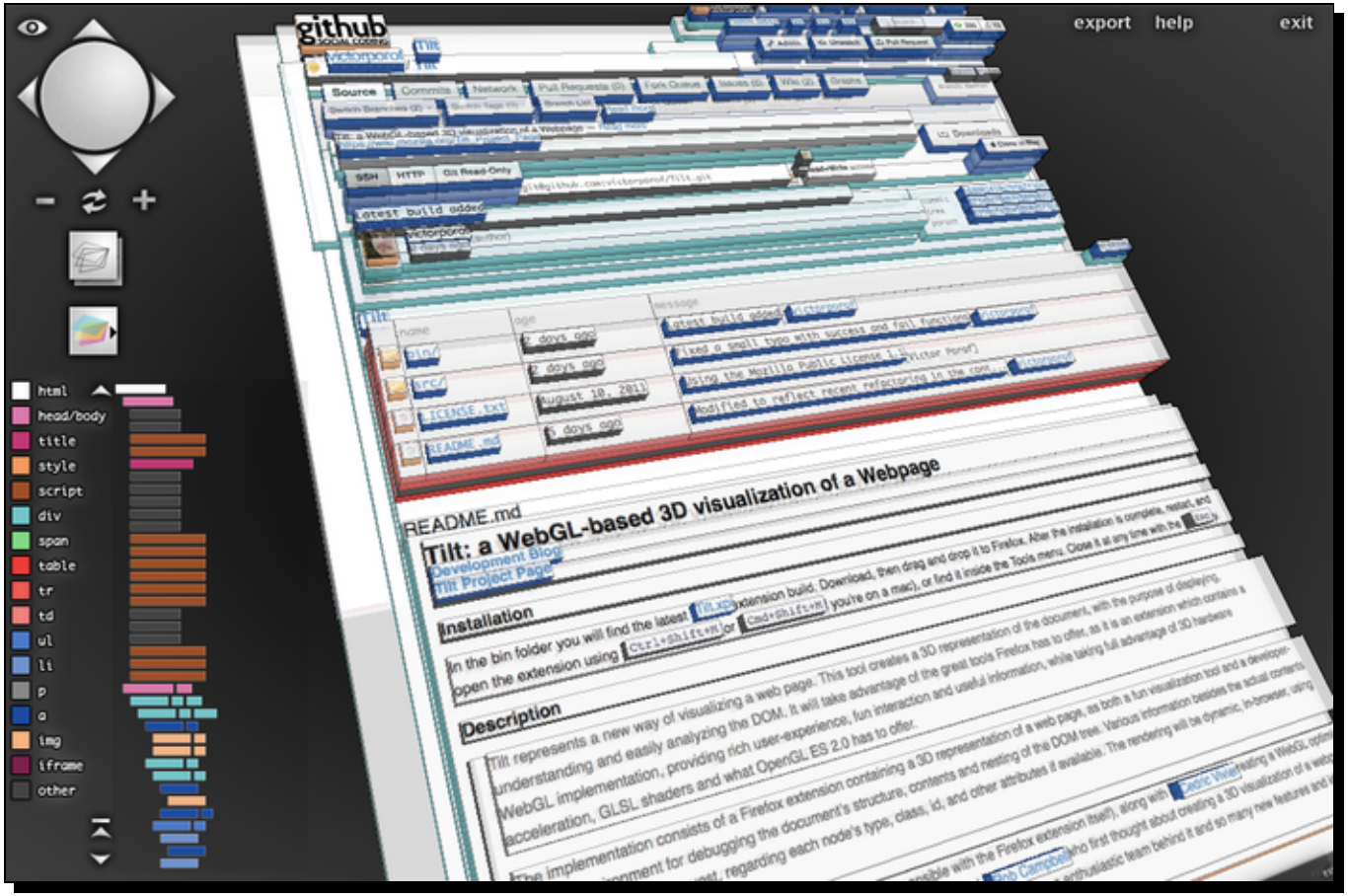
Page Speed Project

- YSlow
  - Staging got a B
  - Production got an F
- Server Recommendations
  - Enable Compression
  - Add Expires Headers
- Suggested Code Changes
  - Optimizing Images (25% savings)
  - Reduce HTTP Requests
    - Combining JavaScript Files
    - CSS Image Sprites
  - JavaScript at Bottom

Google page speed

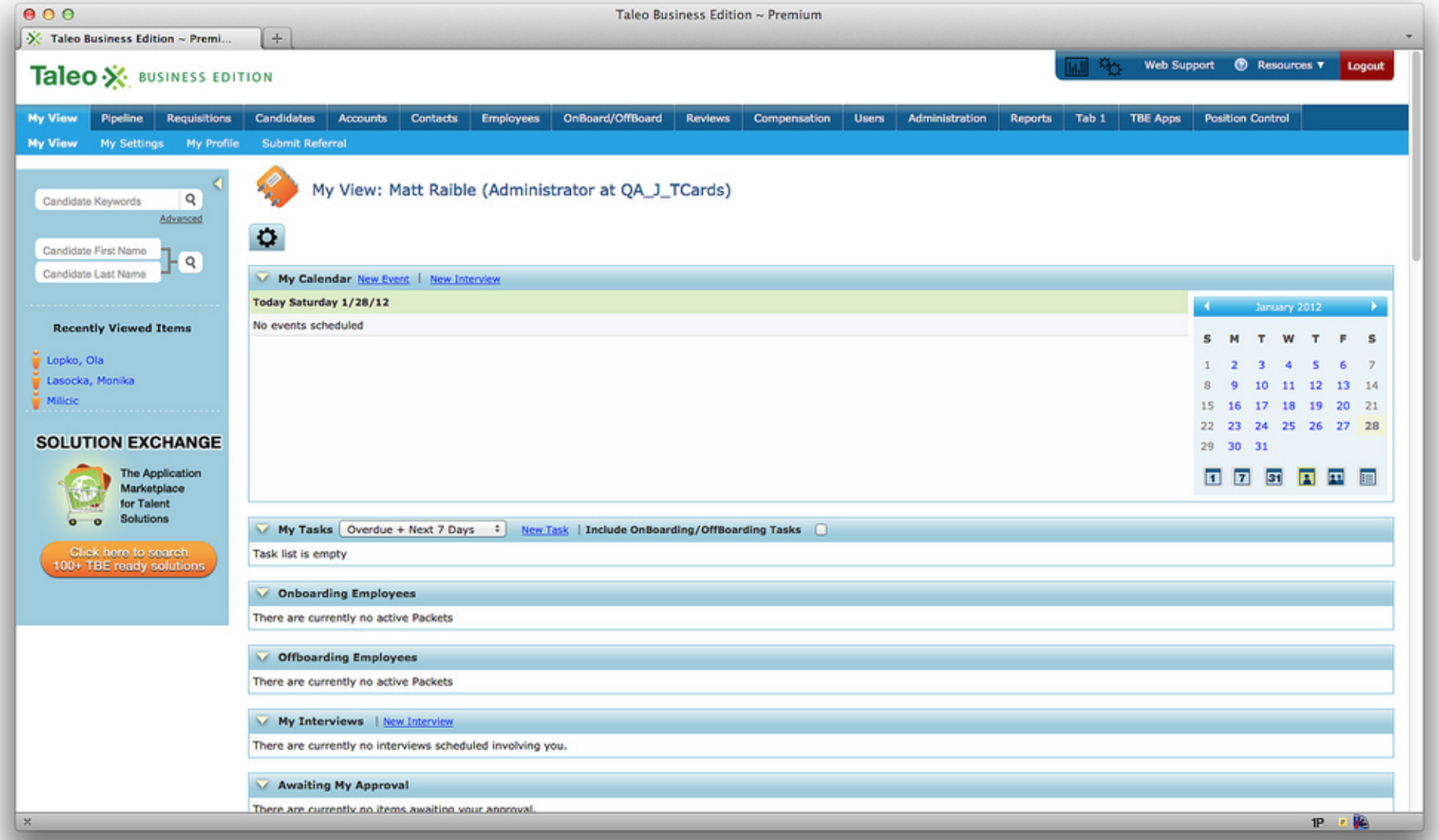


Tilt 3D

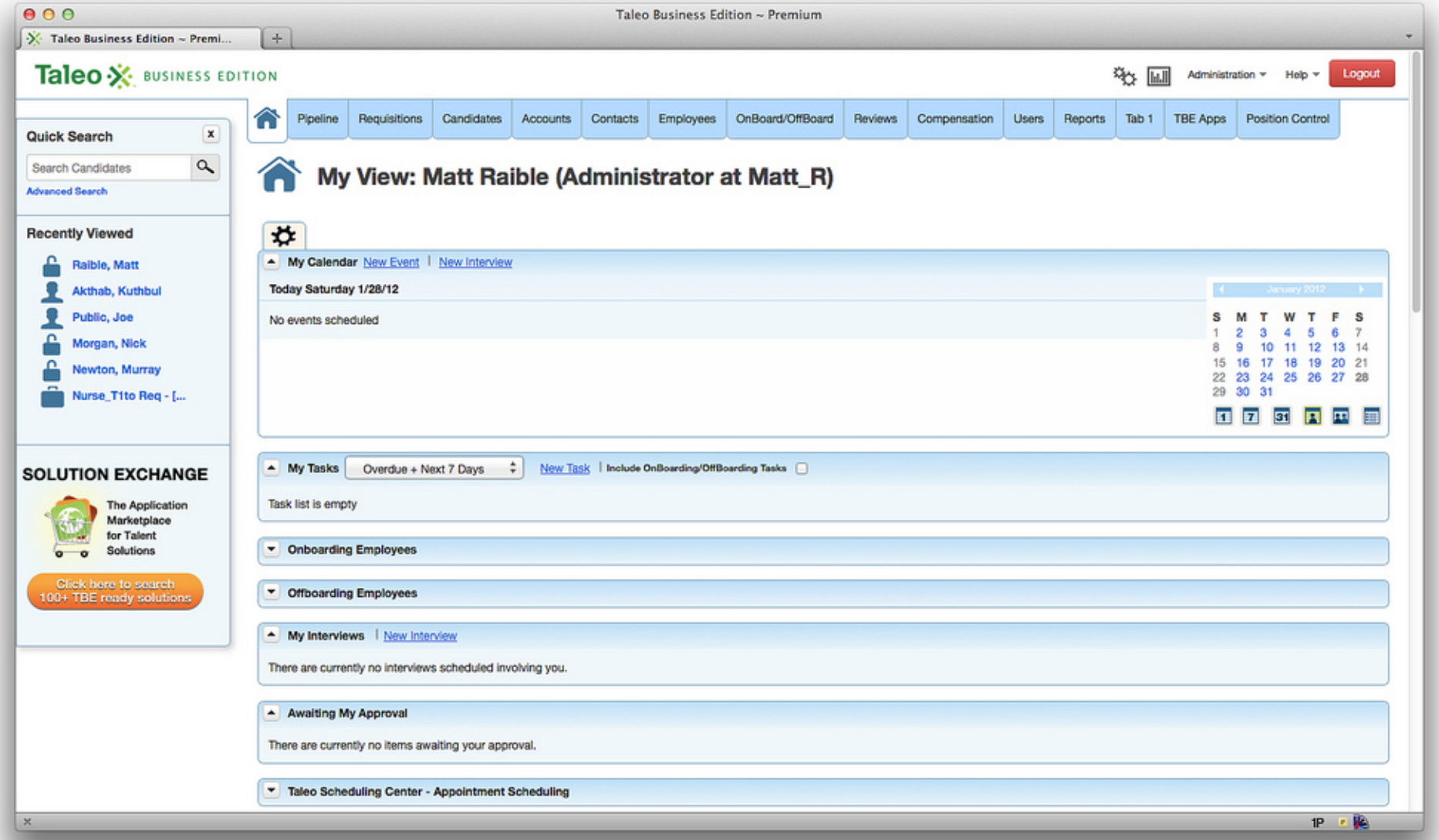


https://addons.mozilla.org/en-us/firefox/addon/tilt/

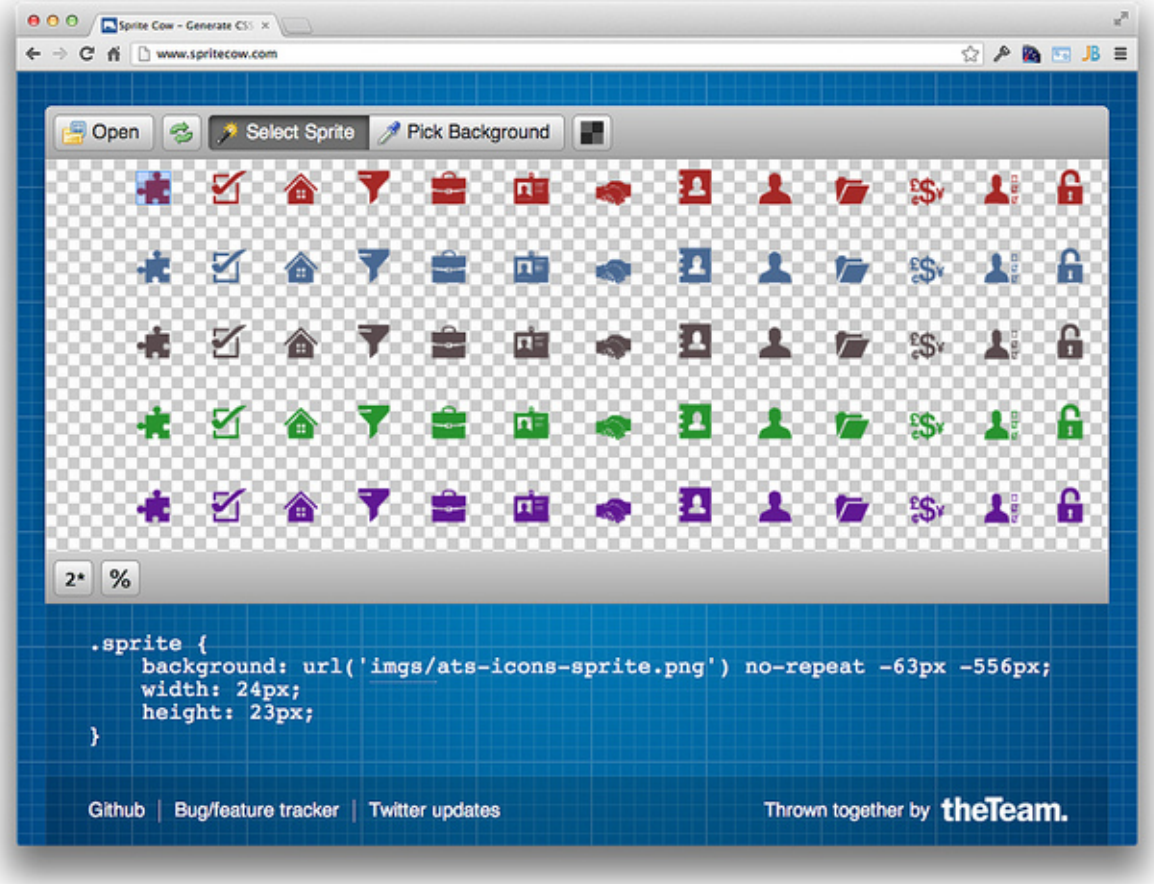
UI Refresh Project



UI Refresh Spike



CSS Image Sprites



http://css-tricks.com/css-sprites/

Themes with LESS

CSS

```
@link: #0087CC;
@linkHover: #0087CC;

@border: #608FAF;

@navbarBackground: #D6E7F2;
@navbarHover: #333333;
@dropdownBackground: #EFF5F9;
@dropdownHover: #D6E7F2;

@sidebarBackground: #EFF5F9;
@headerDarkBackground: #C3EEB7;
@headerMediumBackground: #D6F3CD;
@headerLightBackground: #ffffff;

@iconLargeYPosition: -263px;
@iconYPosition: -718px;
```

Themes with LESS: theme.less

CSS

```
@import '@colors';

a { color: @link }

a:hover { color: @linkHover }

.navbar div > ul > li {
    border: 1px solid @border;
    background-color: @navbarBackground;
    border-top: 0;
}

.navbar-inner {
    border-top: 1px solid @border;
}
```

Themes with LESS: Ant

XML

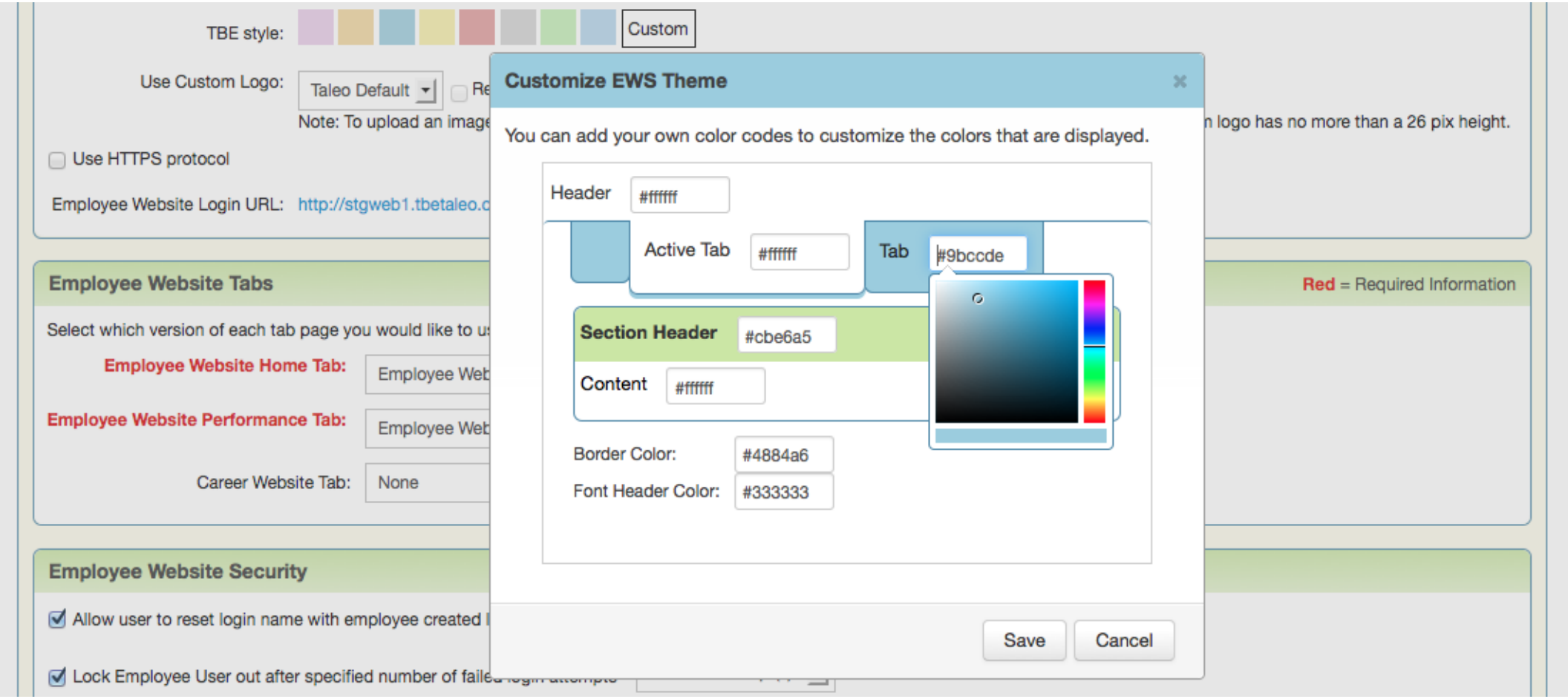
```xml
<macrodef name="create-css">
    <attribute name="theme"/>
    <sequential>
        <echo message="Compiling LESS files :: Creating '@{theme}' Theme"/>
        <copy description="Setup themes for processing" todir="${build}/themes">
            <fileset dir="${basedir}/web/css/UIv9/themes" includes="@{theme}-colors.less"/>
        </copy>
        <concat destfile="${build}/themes/@{theme}.less" eol="dos" force="no">
            <fileset dir="${basedir}/web/css/UIv9/themes">
                <include name="@{theme}-colors.less"/>
                <include name="theme.less"/>
            </fileset>
        </concat>
        <replace file="${build}/themes/@{theme}.less" token="@import '@colors';" value=""/>
        <wro4j processor="lessCss" dir="${build}/themes" todir="${basedir}/web/css/UIv9"
               target-groups="@{theme}"/>
        <delete dir="${basedir}/web/css/UIv9" includes="**-colors.css"/>
    </sequential>
</macrodef>
```

Themes with LESS: Ant

XML

```xml
<macrodef name="wro4j">
    <attribute name="processor"/>
    <attribute name="wro-file" default="wro.xml"/>
    <attribute name="target-groups"/>
    <attribute name="dir"/>
    <attribute name="todir"/>
    <sequential>
        <exec executable="java" failonerror="true">
            <arg value="-jar"/>
            <arg value="../common/lib-ext/wro4j-runner-1.4.4-jar-with-dependencies.jar"/>
            <arg value="--wroFile"/>
            <arg value="@{wro-file}"/>
            <arg value="--contextFolder"/>
            <arg value="@{dir}"/>
            <arg value="--targetGroups"/>
            <arg value="@{target-groups}"/>
            <arg value="--destinationFolder"/>
            <arg value="@{todir}"/>
            <arg value="-c"/>
            <arg value="@{processor}"/>
        </exec>
    </sequential>
</macrodef>
```

Themes with LESS: Customizer



Browser Issues

No rounded corners or gradients in legacy browsers

Used jQuery.IE9Gradius.js for IE9

Had to add class="checkbox" to all checkbox labels

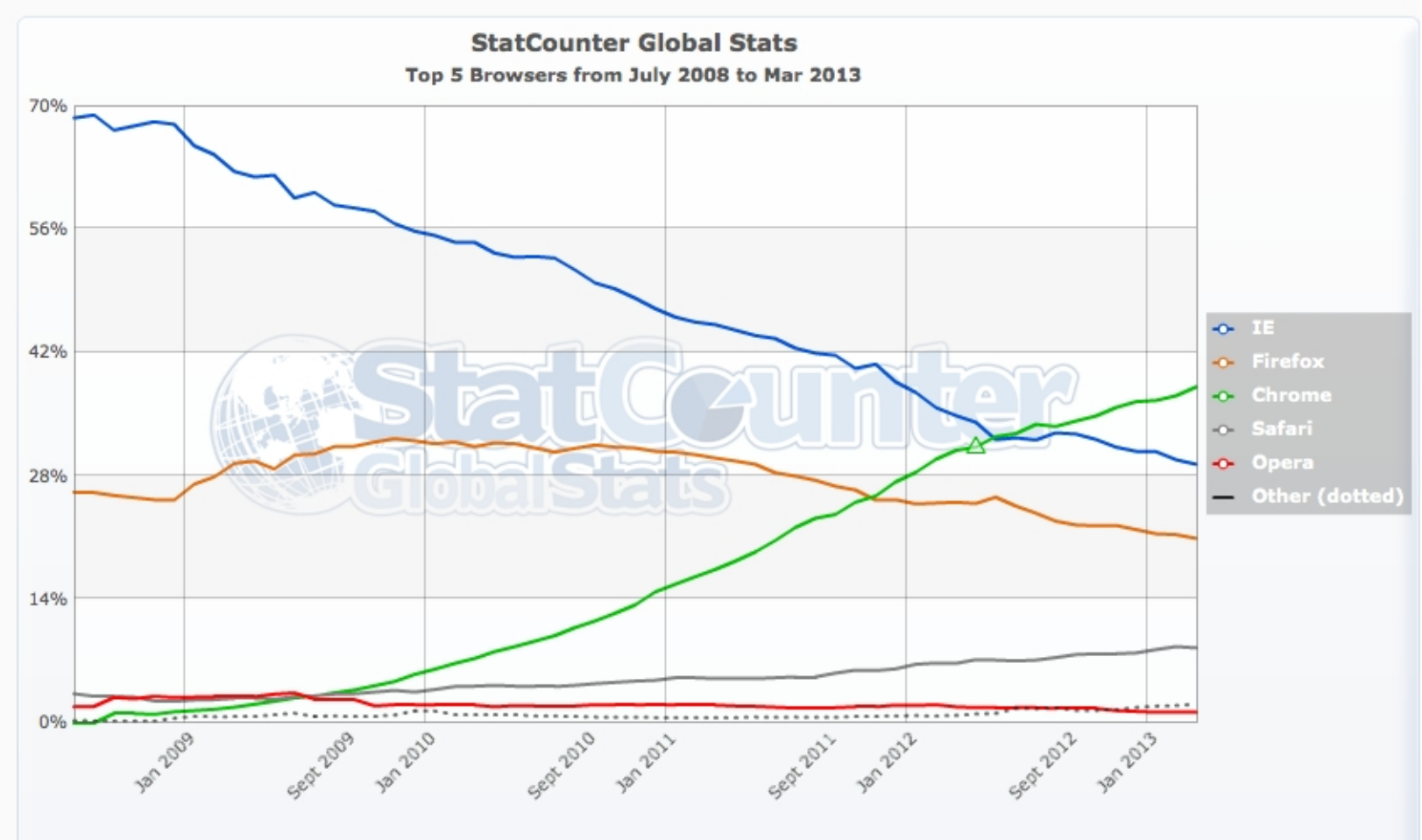
Many, many IE7 Issues: created ie7.css file

- Used `{ zoom: 1; position: relative }` for negative top margins
- Used `table-layout: fixed` so td widths would work
- Used `position: fixed` to fix nav width

Web Development

We face a lot of challenges

Web Browsers

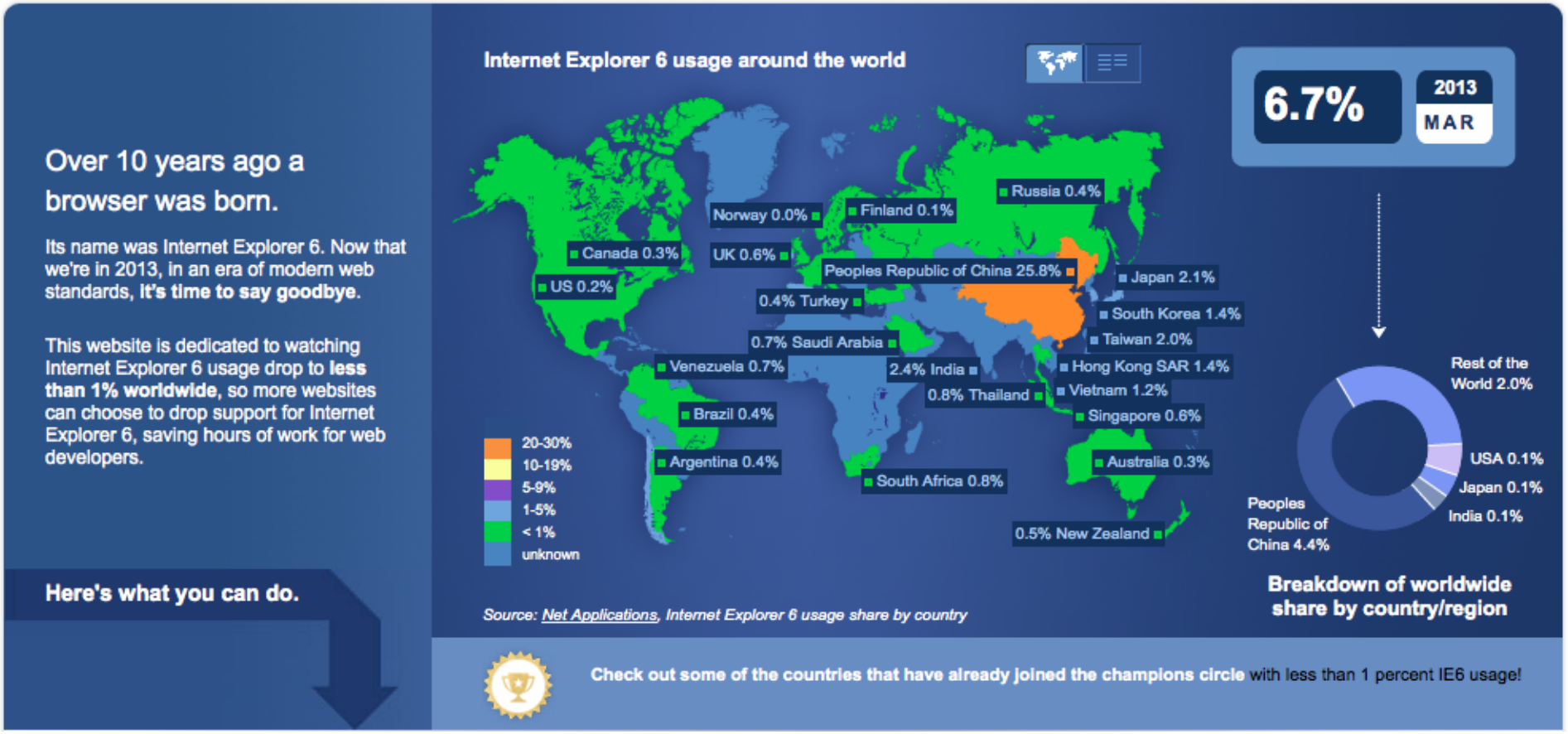


The Internet Explorer 6 Countdown

Moving the world off Internet Explorer 6

Internet Explorer

Download Now



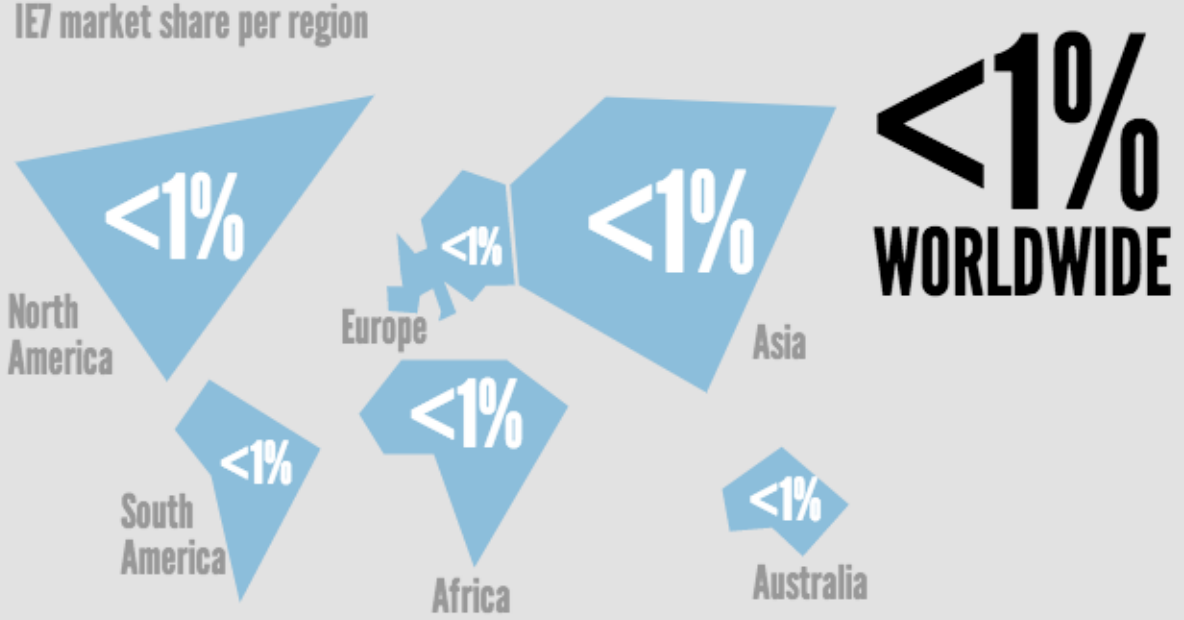
The Internet Explorer 7 Countdown

Moving the world off Internet Explorer 7

Update: We did it! IE7 is below 1% across the world!

Thanks everyone for their part of help. Now let's watch IE8 countdown, IE9 countdown and IE10 countdown too!

This website was dedicated to watching Internet Explorer 7 usage drop to less than 1% worldwide, so more websites could choose to drop support for Internet Explorer 7, saving hours of work for web developers.



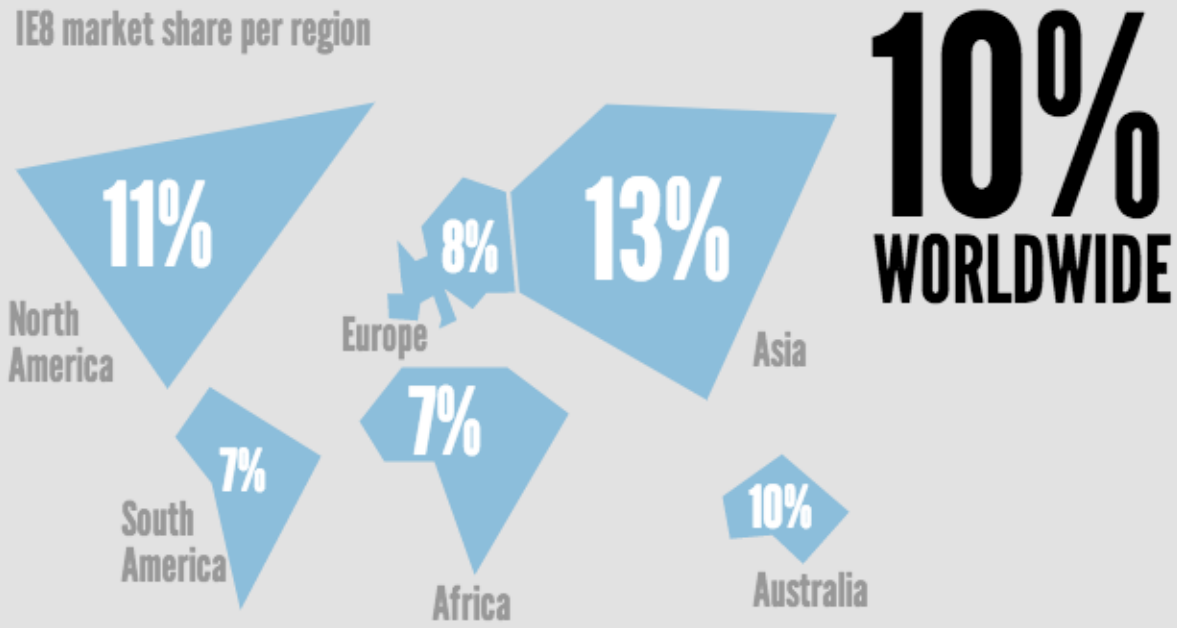
The Internet Explorer 8 Countdown

Moving the world off Internet Explorer 8

5 years ago
another "browser" was born.

Its name was Internet Explorer 8. Now that we're in 2013, in an era of modern web standards, it's time to say goodbye.

This website is dedicated to watching Internet Explorer 8 usage drop to less than 1% worldwide, so more websites can choose to drop support for Internet Explorer 8, saving hours of work for web developers.



Today's Web Developer

HTML5

- Canvas / Semantics/ Storage / File Access / Audio / Video

CSS / 3

- LESS / Sass / @font-face / Border Radius / Box Shadow

JavaScript

- jQuery / Plugins / MVC Frameworks / Server Side JS / CoffeeScript

Server Side Languages

- ColdFusion / Java / Groovy / PHP / Python / .NET / Ruby

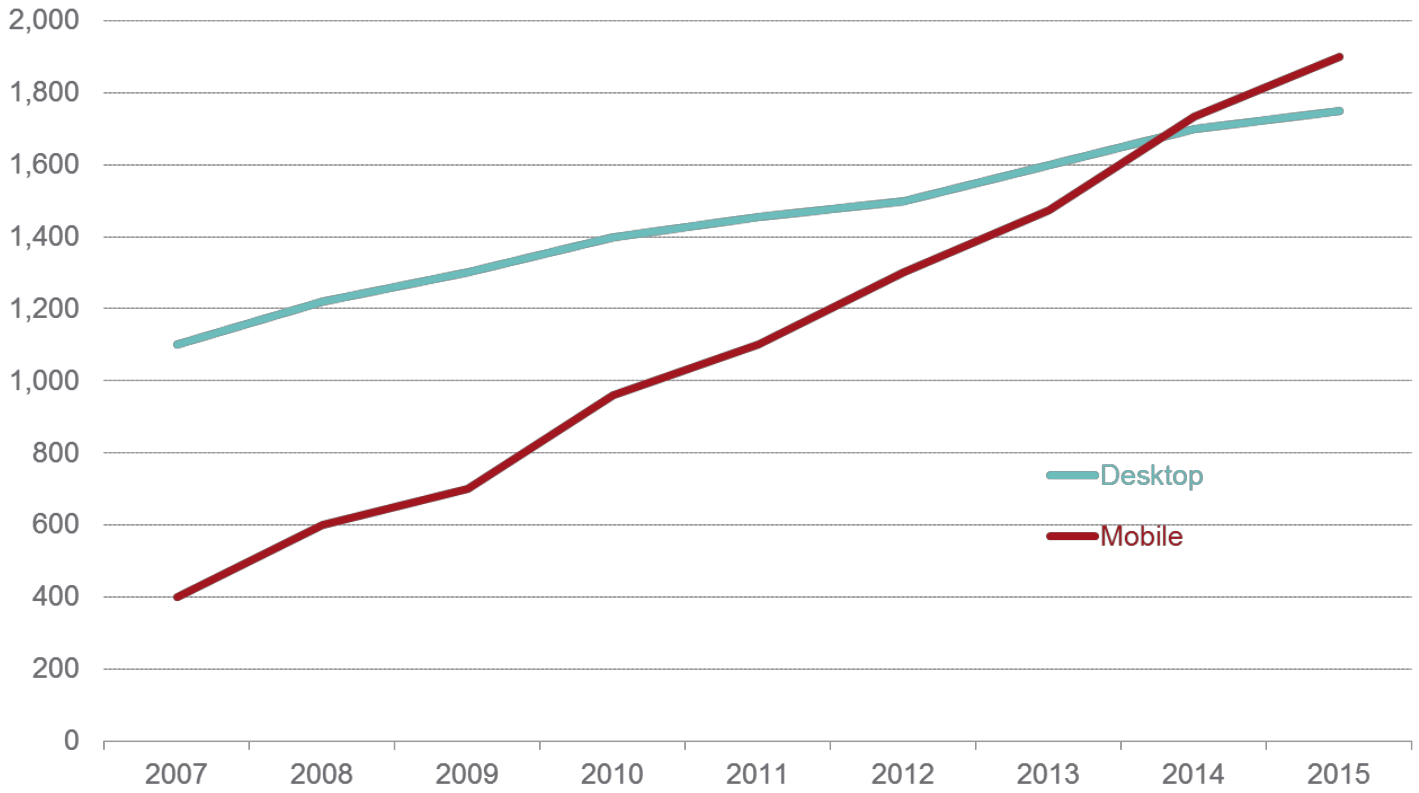
DBMS

- Oracle / SQL Server / MySQL / PostgreSQL

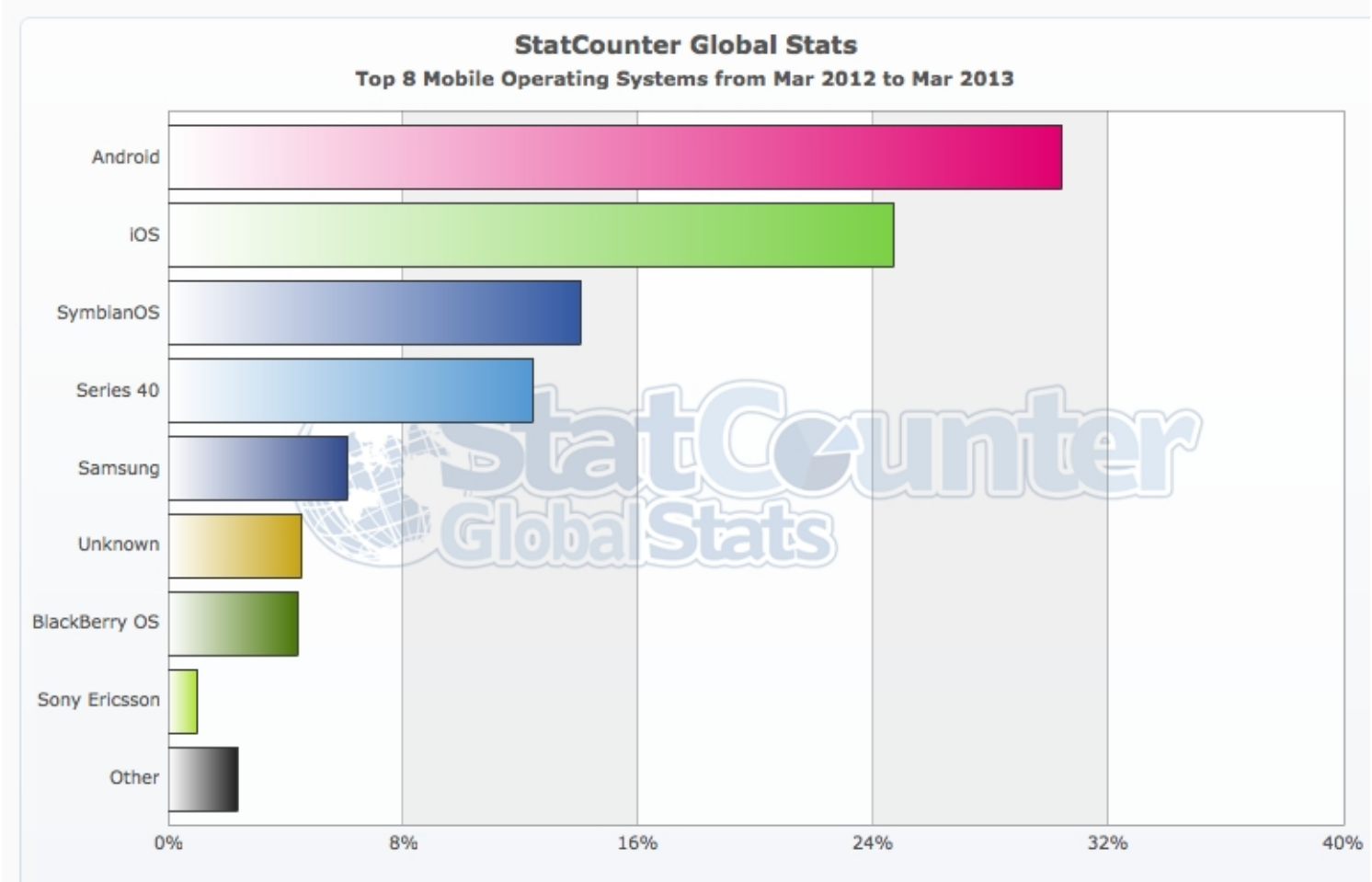
NOSQL Databases

- Cassandra / MongoDB / CouchDB / Amazon Simple DB

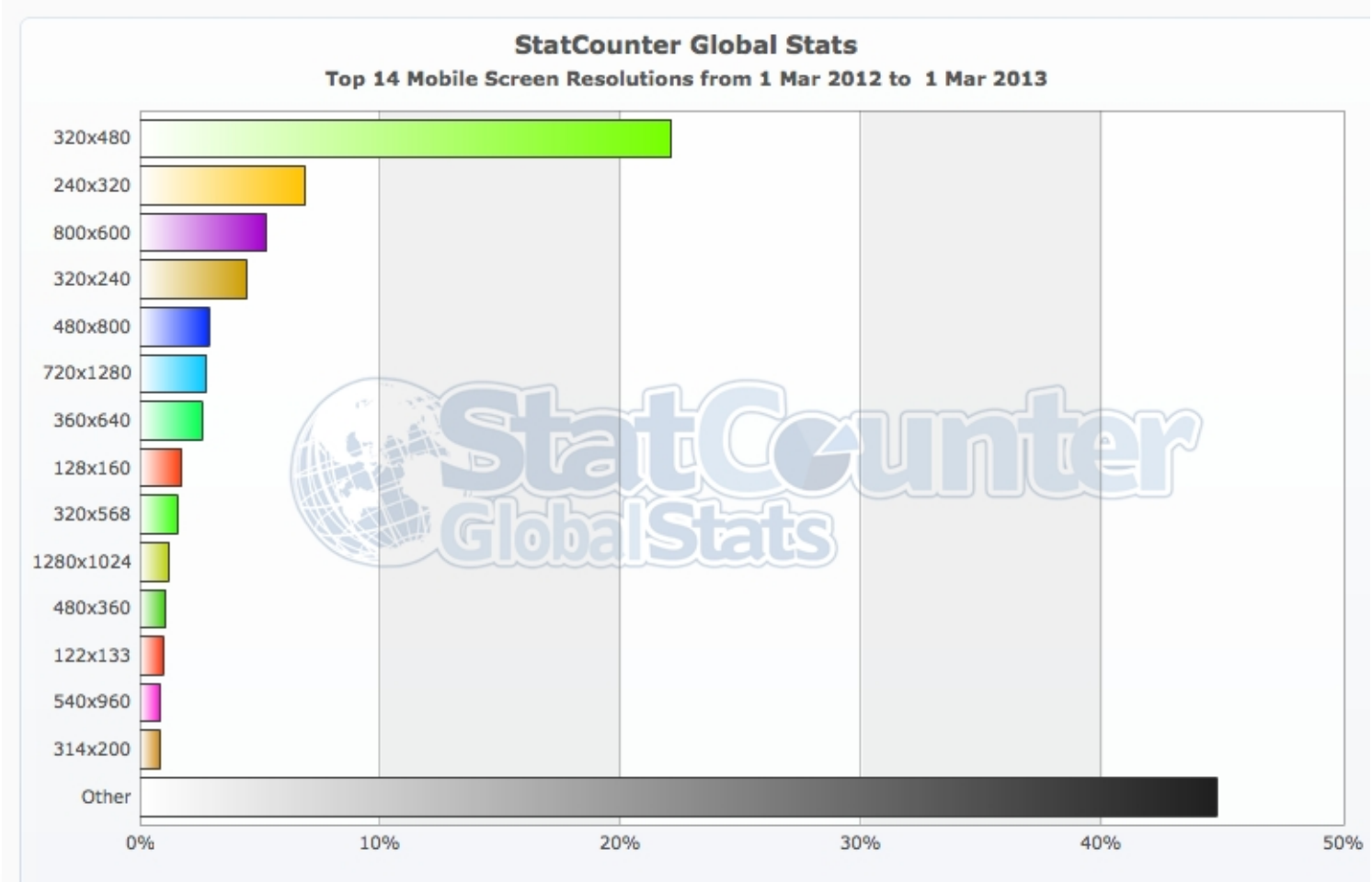
Mobile Traffic will outnumber desktop by 2014



Mobile OS


StatCounter Global Stats
Top 8 Mobile Operating Systems from Mar 2012 to Mar 2013
Android
iOS
SymbianOS
Series 40
Samsung
Unknown
BlackBerry OS
Sony Ericsson
Other
0%
8%
16%
24%
32%
40%
StatCounter
GlobalStats

Mobile Screen Resolutions



“Designing a different interface for every device is impossible.”

Bootstrap

The answer to our problems.

Bootstrap

Sleek, intuitive, and powerful front-end framework for faster and easier web development.

By nerds, for nerds.

Built at Twitter by @mdo and @fat, Bootstrap utilizes LESS CSS, is compiled via Node, and is managed through GitHub to help nerds do awesome stuff on the web.

Made for everyone.

Bootstrap was made to not only look and behave great in the latest desktop browsers (as well as IE7!), but in tablet and smartphone browsers via responsive CSS as well.

Packed with features.

A 12-column responsive grid, dozens of components, javascript plugins, typography, form controls, and even a web-based Customizer to make Bootstrap your own.

What is Bootstrap?

- **Scaffolding**
  - Layouts
  - Grid System
  - Fluid Grid System
  - Responsive Design
- **Base CSS**
  - Typography
  - Tables
  - Forms
  - Buttons
  - Icons
- **Components**
  - Dozens of reusable components
  - Dropdowns
  - Navbar
  - Breadcrumbs
  - Alerts
  - Progress Bars
- **JavaScript Plugins**
  - 13 plugins
  - Modal Dialog Windows
  - Tabs
  - Typeahead
  - Carousel

Reasons To Love Bootstrap

- For all skill levels
- Documentation
- Cross Everything
- Lightweight
- Responsive Design
- 13 JavaScript plugins
- Open Source
- Built on LESS

History of Bootstrap

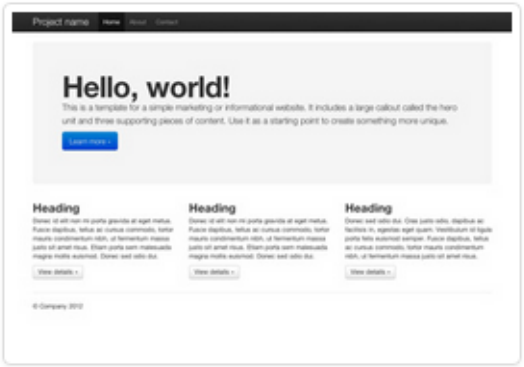
- Early Days of Twitter
  - Mark Otto
  - Jacob Thornton
- Releases
  - v 1.0 - 08/19/2011
  - v 2.0 - 01/31/2012
  - v 2.1.1 - 09/04/2012
  - v 2.3.1 - 03/01/2013
- Popularity
  - @twbootstrap 58.5K Followers
  - GitHub Stats
    - 48K Watchers
    - 14K Forks

Getting Started

Overview of the project, its contents, and how to get started with a simple template.

Getting Started

- Downloading Bootstrap
  - Source
  - Ready To Go
  - Customize
- Requirements
  - HTML5 Doctype
  - Include bootstrap.css
- Starter Templates
  - Basic marketing site
  - Fluid layout
  - Starter template
  - More templates...



Scaffolding

Bootstrap is built on responsive 12-column grids, layouts, and components.

Layouts: Fixed Layout

HTML

```
<body>
  <div class="container">
    ...
  </div>
</body>
```

Layouts: Fluid Layout

HTML

```
<div class="container-fluid">
  <div class="row-fluid">
    <div class="span2">
    </div>
    <div class="span10">
    </div>
  </div>
</div>
```

Grid System

Bootstrap grid system is 12 columns / 940px

Grid adapts to be 724px and 1170px wide depending on your viewport.

Below 767px viewports, the columns become fluid and stack vertically.

HTML

```
<div class="row">
    <div class="span4">...</div>
    <div class="span4">...</div>
    <div class="span4">...</div>
</div>
```

Fluid Grid System

HTML

```
<div class="row-fluid">
    <div class="span4">...</div>
    <div class="span4">...</div>
    <div class="span4">...</div>
</div>
<div class="row-fluid">
    <div class="span6">...</div>
    <div class="span6">...</div>
</div>
<div class="row-fluid">
    <div class="span4">...</div>
    <div class="span4">...</div>
    <div class="span4">...</div>
</div>
```

Responsive Design

Media Queries

Modify the width of column in our grid

Stack elements instead of float wherever necessary

Resize headings and text to be more appropriate for devices

Responsive Design

HTML

```
<!doctype html>
<html>
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Bootstrap: Getting Started</title>
    <link rel="stylesheet" type="text/css" href="bootstrap/css/bootstrap.min.css">
    <link rel="stylesheet" type="text/css" href="assets/css/bootstrap-responsive.css" >
</head>
<body>

<h1>Responsive Bootstrap</h1>

</body>
</html>
```

Demo Time

http://localhost:8000/demos/responsive/

Base CSS

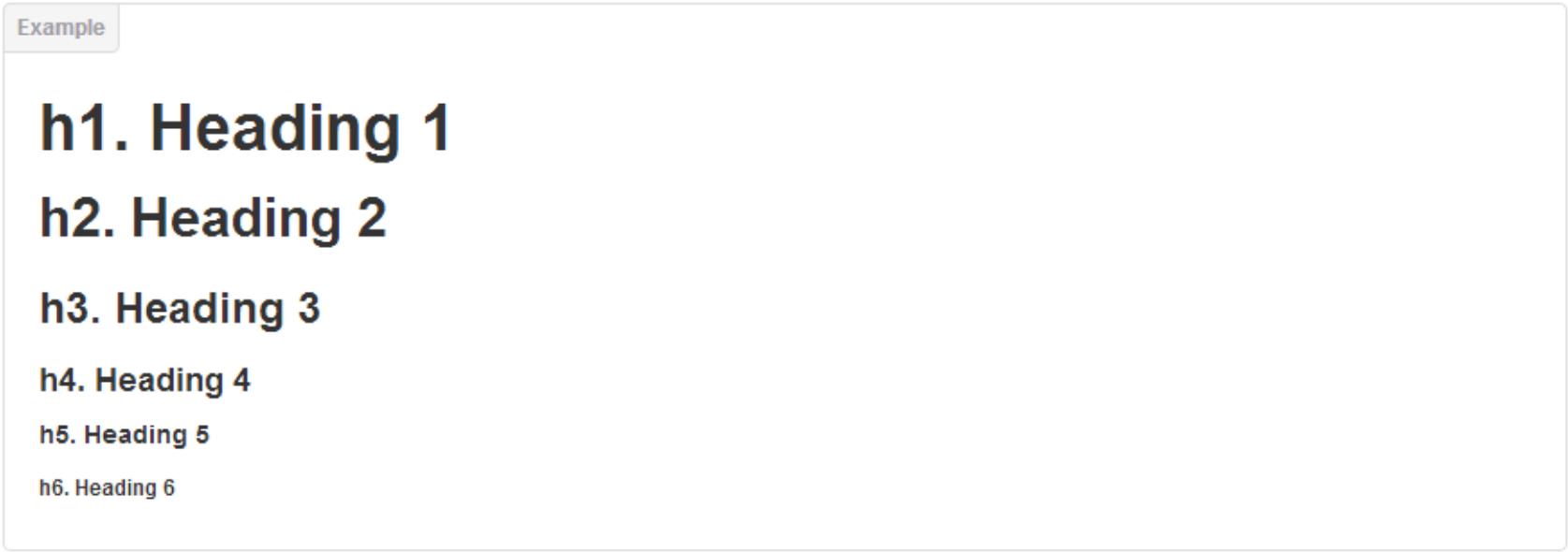
Fundamental HTML elements styled and enhanced with extensible classes.

Typography: Headings

HTML

```
<h1>h1. Heading 1</h1>
<h2>h2. Heading 2</h2>
<h3>h3. Heading 3</h3>
<h4>h4. Heading 4</h4>
<h5>h5. Heading 5</h5>
<h6>h6. Heading 6</h6>
```

Example



Typography: Body Copy

```HTML
<p>Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridicu
Donec ullamcorper nulla non metus auctor fringilla. Duis mollis, est non commodo luctus, nisi
erat porttitor ligula, eget lacinia odio sem nec elit. Donec ullamcorper nulla non metus
auctor fringilla.</p>
```

Example

Nullam quis risus eget urna mollis ornare vel eu leo. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Nullam id dolor id nibh ultricies vehicula.

Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec ullamcorper nulla non metus auctor fringilla. Duis mollis, est non commodo luctus, nisi erat porttitor ligula, eget lacinia odio sem nec elit. Donec ullamcorper nulla non metus auctor fringilla.

Maecenas sed diam eget risus varius blandit sit amet non magna. Donec id elit non mi porta gravida at eget metus. Duis mollis, est non commodo luctus, nisi erat porttitor ligula, eget lacinia odio sem nec elit.

Typography: Blockquote

HTML

```
<blockquote>
  <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer posuere erat a ante.</p>
  <small>Someone famous <cite title="Source Title">Source Title</cite></small>
</blockquote>
```

Example

> Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer posuere erat a ante.
>
> — Someone famous in Source Title

Typography: Emphasis classes

```html
<p class="muted">Fusce dapibus, tellus ac cursus commodo, tortor mauris nibh.</p>
<p class="text-warning">Etiam porta sem malesuada magna mollis euismod.</p>
<p class="text-error">Donec ullamcorper nulla non metus auctor fringilla.</p>
<p class="text-info">Aenean eu leo quam. Pellentesque ornare sem lacinia quam venenatis.</p>
<p class="text-success">Duis mollis, est non commodo luctus, nisi erat porttitor ligula.</p>
```

HTML

Example

Fusce dapibus, tellus ac cursus commodo, tortor mauris nibh.

Etiam porta sem malesuada magna mollis euismod.

Donec ullamcorper nulla non metus auctor fringilla.

Aenean eu leo quam. Pellentesque ornare sem lacinia quam venenatis.

Duis mollis, est non commodo luctus, nisi erat porttitor ligula.

Typography: Lists

HTML

```
<ul>
<li>Lorem ipsum dolor sit amet</li>
...
</ul>
```

Example

- Lorem ipsum dolor sit amet
- Consectetur adipiscing elit
- Integer molestie lorem at massa
- Facilisis in pretium nisl aliquet
- Nulla volutpat aliquam velit
  - Phasellus iaculis neque
  - Purus sodales ultricies
  - Vestibulum laoreet porttitor sem
  - Ac tristique libero volutpat at
- Faucibus porta lacus fringilla vel
- Aenean sit amet erat nunc
- Eget porttitor lorem

Tables: Default Style

```html
<table class="table">
…
</table>
```

HTML

Example

#	First Name	Last Name	Username
1	Mark	Otto	@mdo
2	Jacob	Thornton	@fat
3	Larry	the Bird	@twitter

Tables: Striped

HTML

```
<table class="table table-striped">
…
</table>
```

Example

#	First Name	Last Name	Username
1	Mark	Otto	@mdo
2	Jacob	Thornton	@fat
3	Larry	the Bird	@twitter

Tables: Bordered

```
<table class="table table-bordered">
...
</table>
```

HTML

Example

#	First Name	Last Name	Username
1	Mark	Otto	@mdo
	Mark	Otto	@TwBootstrap
2	Jacob	Thornton	@fat
3	Larry the Bird		@twitter

Tables: Hover

```html
<table class="table table-hover">
...
</table>
```

HTML

Example

#	First Name	Last Name	Username
1	Mark	Otto	@mdo
2	Jacob	Thornton	@fat
3	Larry the Bird		@twitter

Tables: Condensed

```html
<table class="table table-condensed">
…
</table>
```

HTML

Example

#	First Name	Last Name	Username
1	Mark	Otto	@mdo
2	Jacob	Thornton	@fat
3	Larry the Bird		@twitter

Tables: Optional Row Classes

Class	Description
`.success`	Indicates a successful or positive action.
`.error`	Indicates a dangerous or potentially negative action.
`.warning`	Indicates a warning that might need attention.
`.info`	Used as an alternative to the default styles.

Example

#	Product	Payment Taken	Status
1	TB - Monthly	01/04/2012	Approved
2	TB - Monthly	02/04/2012	Declined
3	TB - Monthly	03/04/2012	Pending
4	TB - Monthly	04/04/2012	Call in to confirm

Forms

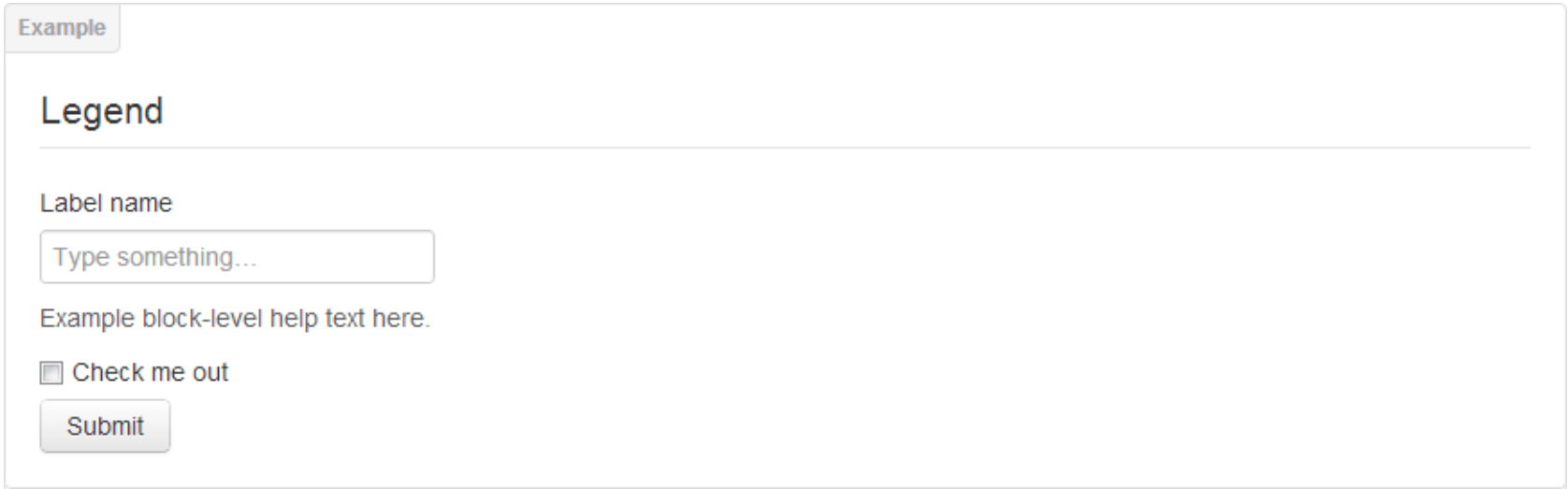
- Four Layouts
  - Vertical (default)
  - Search
  - Inline
  - Horizontal
- Form Controls
  - Input
  - Text Area
  - Checkboxes & Radio
  - Select
- Look & Feel
  - Relative Sizing (small/med/large/xlarge)
  - Grid Sizing (span1/span2/span3/etc)

Forms: Default Styles

HTML

```
<form>
    <legend>Legend</legend>
    <label>Label name</label>
    <input type="text" placeholder="Type something…">
    <span class="help-block">Example block-level help text here.</span>
    <label class="checkbox">
        <input type="checkbox"> Check me out
    </label>
    <button type="submit" class="btn">Submit</button>
</form>
```

Example

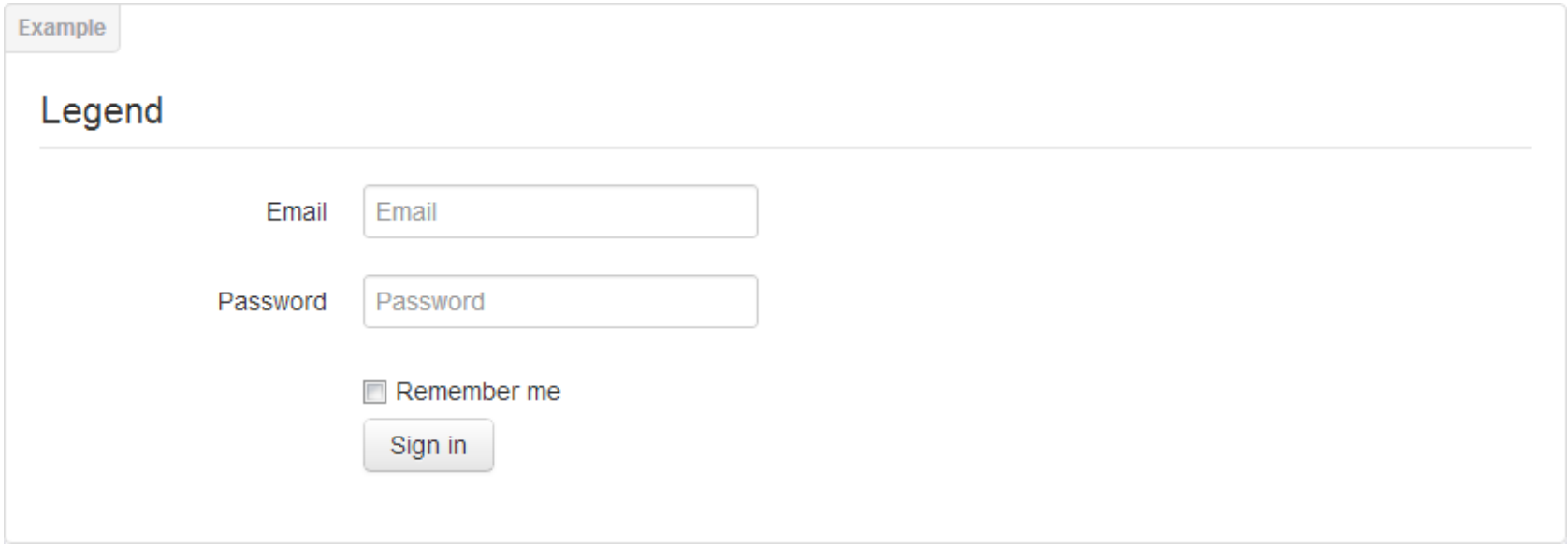


Forms: Horizontal

HTML

```
<form class="form-horizontal">
    <div class="control-group">
        <label class="control-label" for="inputEmail">Email</label>
        <div class="controls">
          <input type="text" id="inputEmail" placeholder="Email">
        </div>
    </div>
    ...
</form>
```

Example

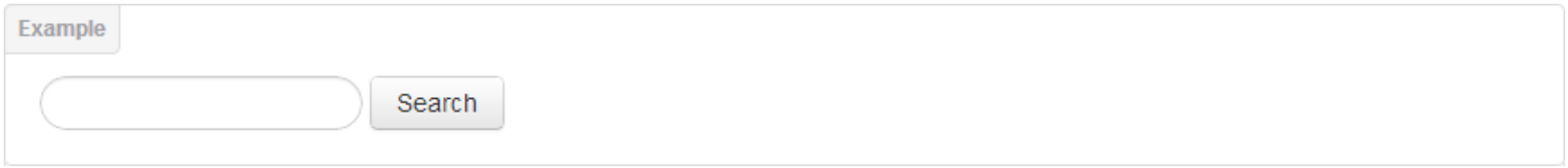


Search Form

HTML

```
<form class="form-search">
    <input type="text" class="input-medium search-query">
    <button type="submit" class="btn">Search</button>
</form>
```

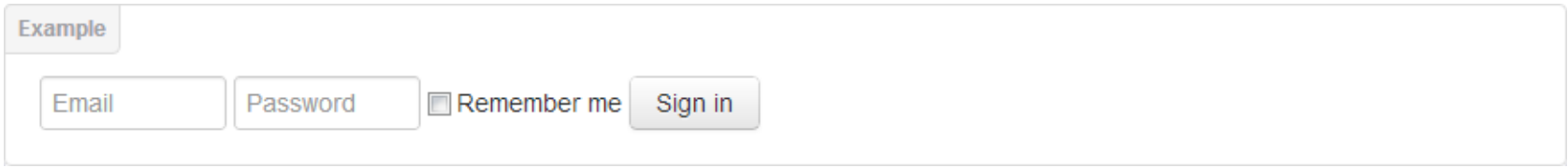
Example



Inline Form

HTML

```
<form class="form-inline">
    <input type="text" class="input-small" placeholder="Email">
    <input type="password" class="input-small" placeholder="Password">
    <label class="checkbox">
        <input type="checkbox"> Remember me
    </label>
    <button type="submit" class="btn">Sign in</button>
</form>
```

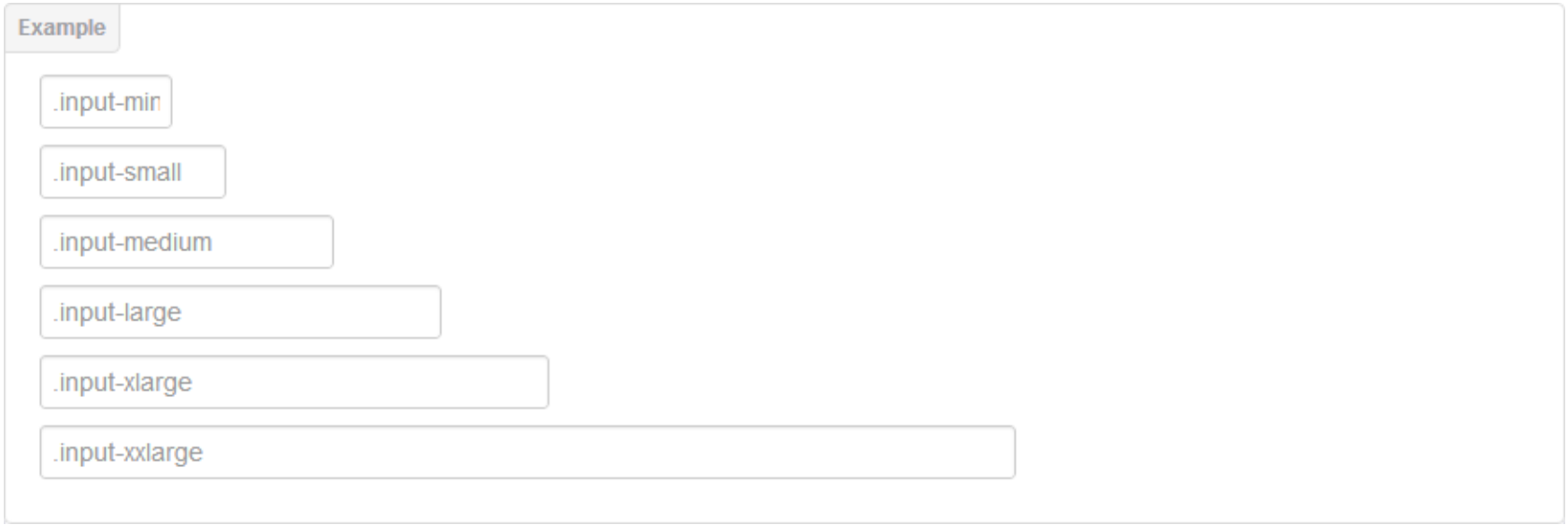


Form Controls: Relative Sizing

HTML

```
<input class="input-mini" type="text" placeholder=".input-mini">
<input class="input-small" type="text" placeholder=".input-small">
<input class="input-medium" type="text" placeholder=".input-medium">
<input class="input-large" type="text" placeholder=".input-large">
<input class="input-xlarge" type="text" placeholder=".input-xlarge">
<input class="input-xxlarge" type="text" placeholder=".input-xxlarge">
```

Example

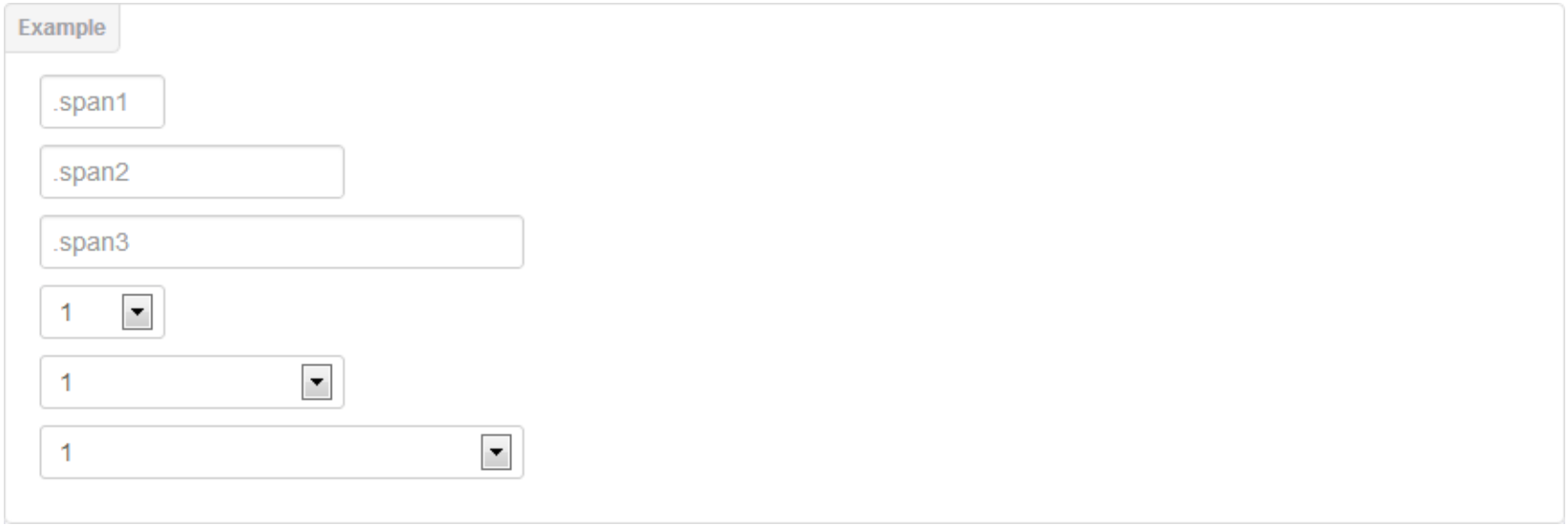


Form Controls: Grid Sizing

HTML

```
<input class="span1" type="text" placeholder=".span1">
<input class="span2" type="text" placeholder=".span2">
<input class="span3" type="text" placeholder=".span3">
<select class="span1"></select>
<select class="span2"></select>
<select class="span3"></select>
```

Example



Buttons

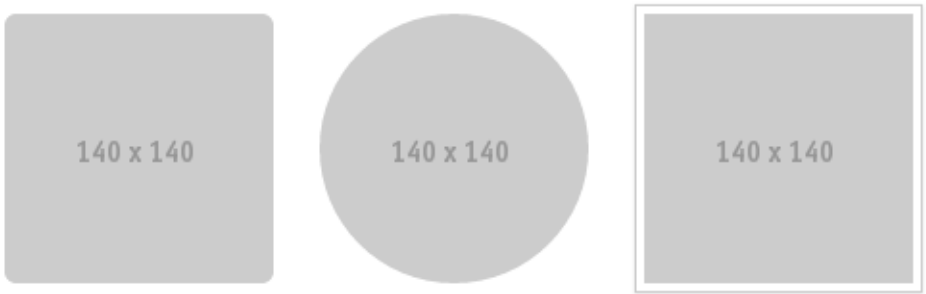
Button	class=""	Description
Default	`btn`	Standard gray button with gradient
Primary	`btn btn-primary`	Provides extra visual weight and identifies the primary action in a set of buttons
Info	`btn btn-info`	Used as an alternative to the default styles
Success	`btn btn-success`	Indicates a successful or positive action
Warning	`btn btn-warning`	Indicates caution should be taken with this action
Danger	`btn btn-danger`	Indicates a dangerous or potentially negative action
Inverse	`btn btn-inverse`	Alternate dark gray button, not tied to a semantic action or use
Link	`btn btn-link`	Deemphasize a button by making it look like a link while maintaining button behavior

Images

HTML

```
<img src="..." class="img-rounded">
<img src="..." class="img-circle">
<img src="..." class="img-polaroid">
```

Example



.img-rounded and .img-circle do not work in IE7-8 due to lack of border-radius support.

Image Placeholders http://placehold.it/

Icons

HTML

```
<i class="icon-search"></i>
<i class="icon-search icon-white"></i>

<a class="btn btn-primary" href="#"><i class="icon-user icon-white"></i> User</a>
<a class="btn btn-small" href="#"><i class="icon-star"></i></a>

<span class="add-on"><i class="icon-envelope"></i></span>
<input class="span2" id="inputIcon" type="text">
```

icon-glass	icon-music	icon-search	icon-envelope
icon-heart	icon-star	icon-star-empty	icon-user
icon-film	icon-th-large	icon-th	icon-th-list
icon-ok	icon-remove	icon-zoom-in	icon-zoom-out
icon-off	icon-signal	icon-cog	icon-trash
icon-home	icon-file	icon-time	icon-road

Full List of 140 Icons http://twitter.github.com/bootstrap/base-css.html#icons

Compressed using ImageOptim http://imageoptim.com/

Components

Dozens of reusable components built to provide navigation, alerts, popovers, and more.

Components

- Dropdowns
- Button groups
- Button dropdowns
- Navs
- Navbar
- Breadcrumbs
- Pagination
- Labels and badges
- Typography
- Thumbnails
- Alerts
- Progress bars
- Misc

Components: Button Group

HTML

```
<div class="btn-group">
    <button class="btn">1</button>
    <button class="btn">2</button>
    <button class="btn">3</button>
</div>
```

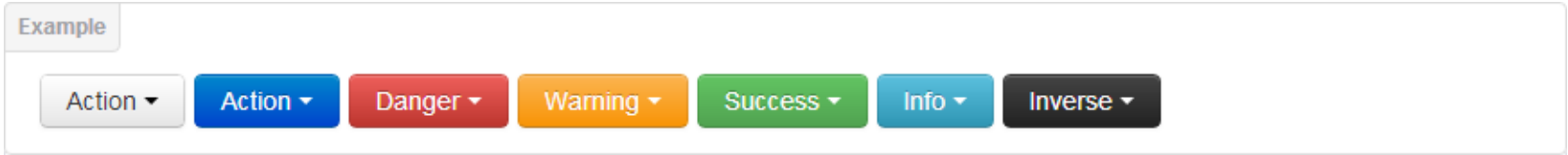
Example

Components: Button dropdown

HTML

```
<div class="btn-group">
<a class="btn dropdown-toggle" data-toggle="dropdown" href="#">
    Action
    <span class="caret"></span>
</a>
<ul class="dropdown-menu">
    <!-- dropdown menu links -->
</ul>
</div>
```

Example

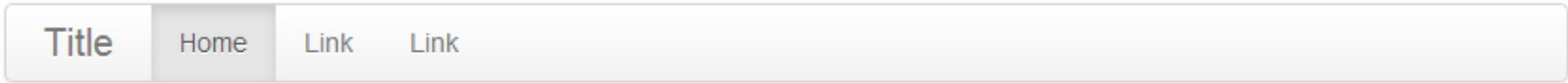


Components: Navbar

HTML

```
<div class="navbar">
  <div class="navbar-inner">
    <a class="brand" href="#">Title</a>
    <ul class="nav">
      <li class="active"><a href="#">Home</a></li>
      <li><a href="#">Link</a></li>
      <li><a href="#">Link</a></li>
    </ul>
  </div>
</div>
```

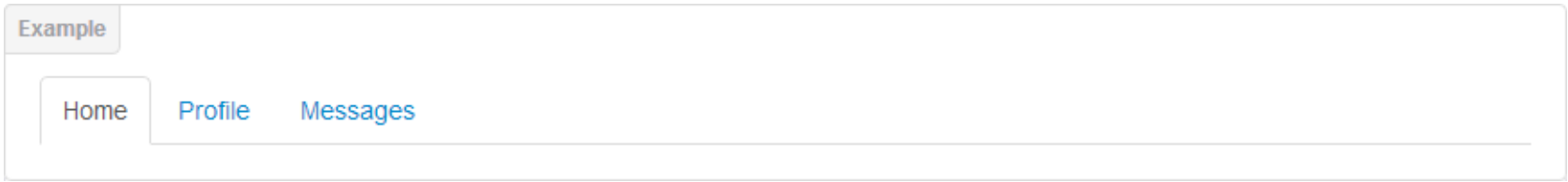
Example



Components: Tabs

HTML

```
<ul class="nav nav-tabs">
<li class="active">
    <a href="#">Home</a>
</li>
<li><a href="#">...</a></li>
<li><a href="#">...</a></li>
</ul>
```



Components: Breadcrumbs

HTML

```
<ul class="breadcrumb">
    <li><a href="#">Home</a> <span class="divider">/</span></li>
    <li><a href="#">Library</a> <span class="divider">/</span></li>
    <li class="active">Data</li>
</ul>
```

Example

Home

Home / Library

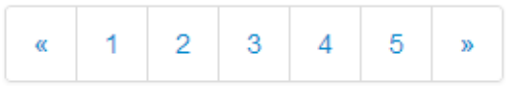
Home / Library / Data

Components: Pagination

HTML

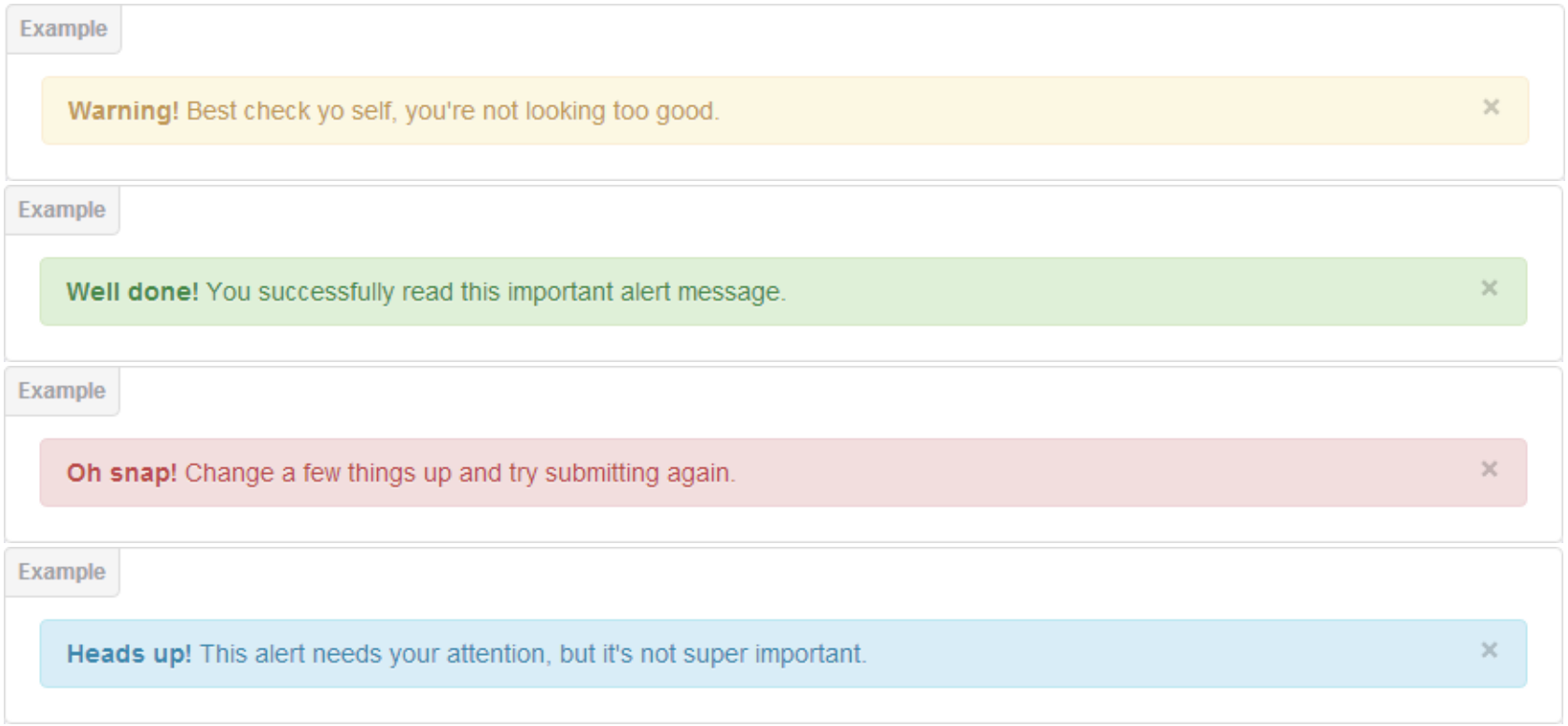
```
<div class="pagination">
    <ul>
        <li><a href="#">Prev</a></li>
        <li><a href="#">1</a></li>
        <li><a href="#">2</a></li>
        <li><a href="#">3</a></li>
        <li><a href="#">4</a></li>
        <li><a href="#">Next</a></li>
    </ul>
</div>
```

Example



Components: Alerts

```HTML
<div class="alert alert-success">
</div>
```



JavaScript for Bootstrap

Bring Bootstrap's components to life—now with 13 custom jQuery plugins.

JavaScript Plugins: Overview

All plugins require the latest version of jQuery

What Plugins do you need?

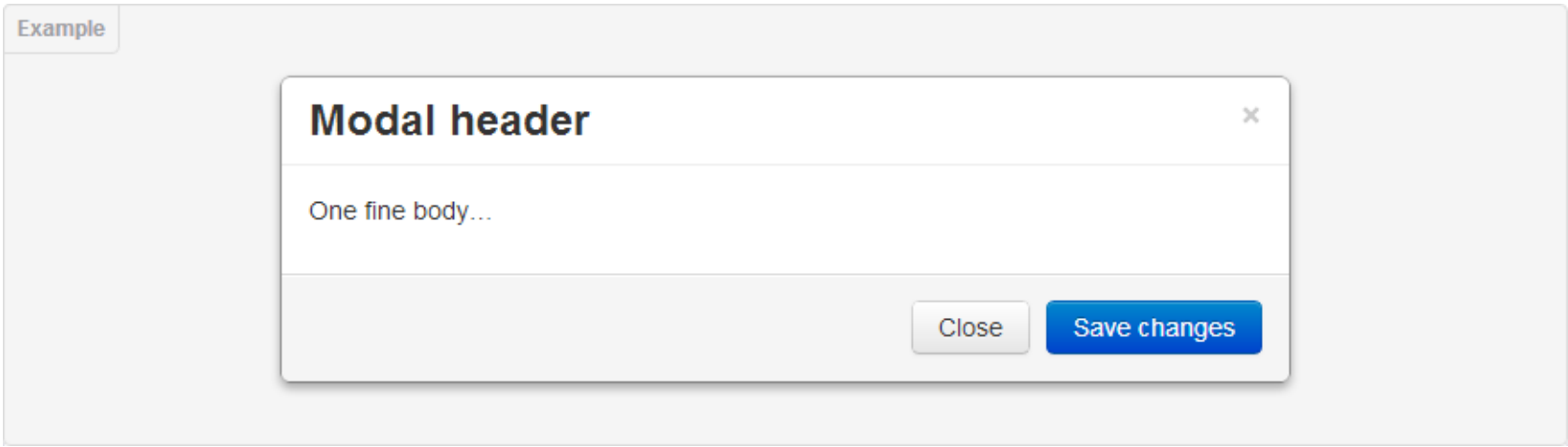
- bootstrap.min.js (all)
- https://github.com/twitter/bootstrap/tree/master/js

Plugins

- Transitions
- Modal
- Button / Dropdown
- Tab
- Tooltip / Popover
- Alert
- Collapse
- Carousel
- Typeahead
- Scrollspy / Affix

“You can use all Bootstrap plugins purely through the markup API without writing a single line of JavaScript.”

JavaScript Plugins: Modals

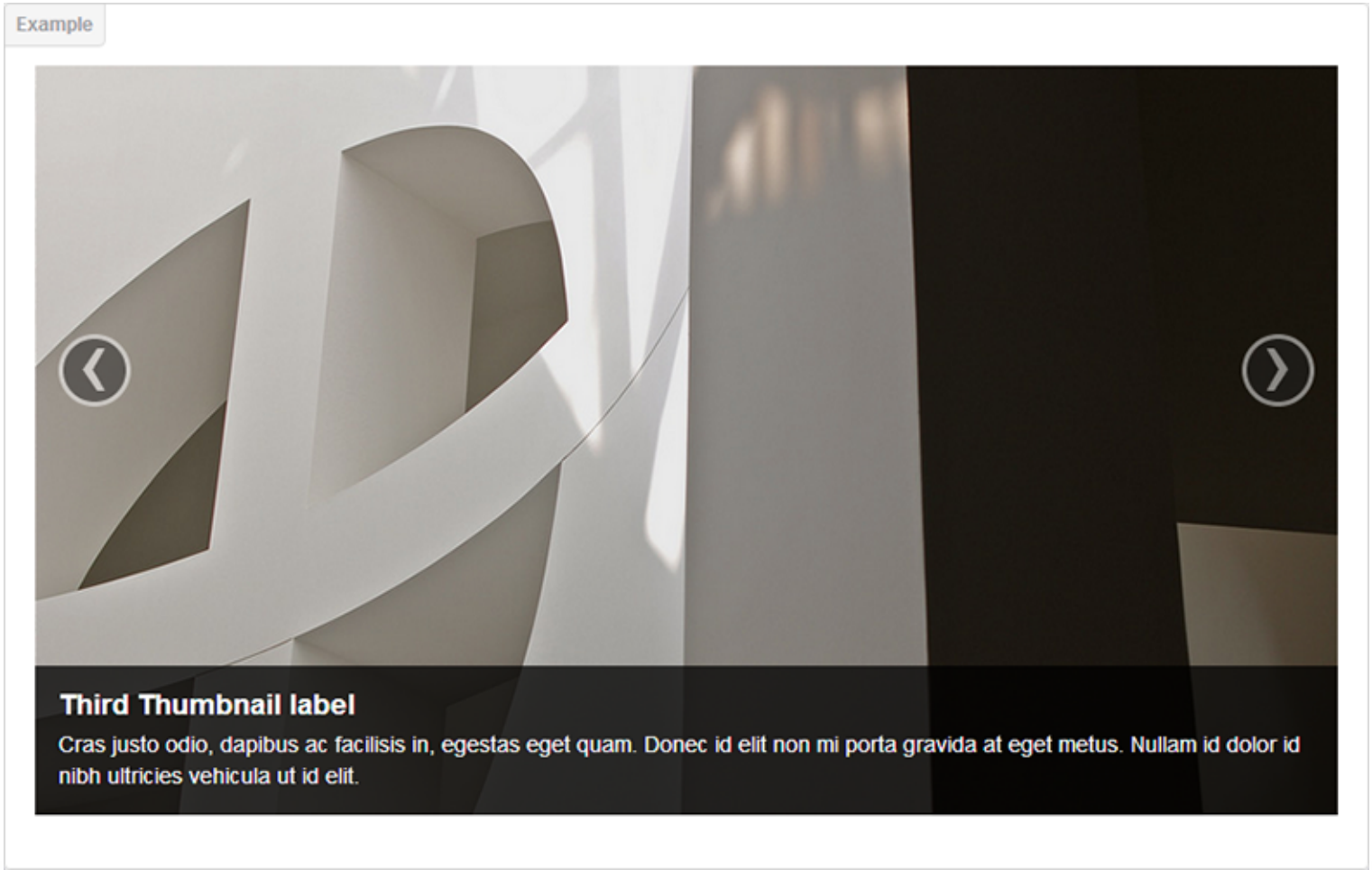


JavaScript Plugins: Modals

HTML

```
<div class="modal hide fade">
    <div class="modal-header">
        <button type="button" class="close" data-dismiss="modal" aria-hidden="true">&times;</bu
        <h3>Modal header</h3>
    </div>
    <div class="modal-body">
        <p>One fine body…</p>
    </div>
    <div class="modal-footer">
        <a href="#" class="btn">Close</a>
        <a href="#" class="btn btn-primary">Save changes</a>
    </div>
</div>
```

JavaScript Plugins: Carousel

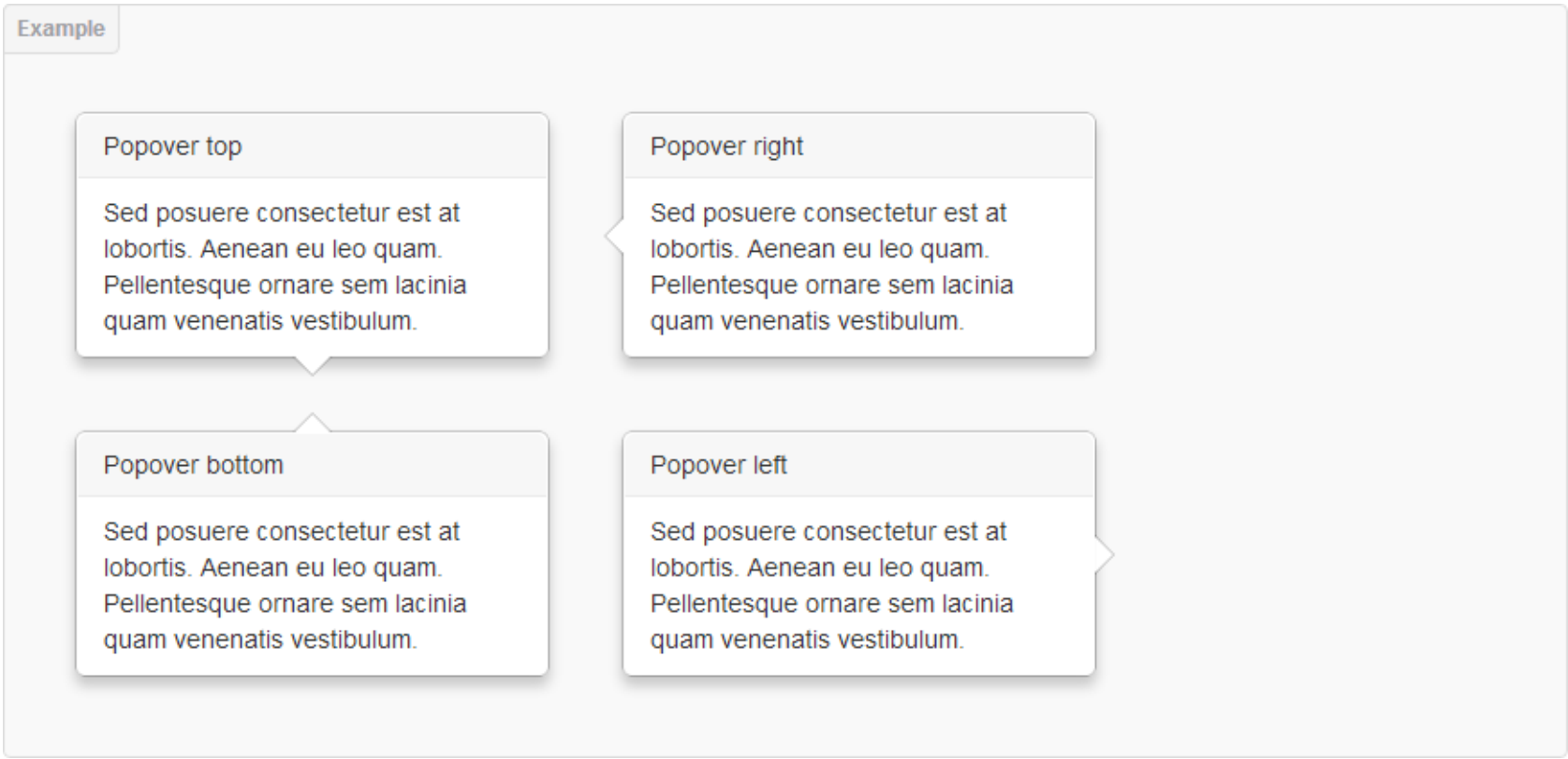


JavaScript Plugins: Carousel

HTML

```
<div id="myCarousel" class="carousel slide">
    <!-- Carousel items -->
    <div class="carousel-inner">
        <div class="active item">…</div>
        <div class="item">…</div>
        <div class="item">…</div>
    </div>
    <!-- Carousel nav -->
    <a class="carousel-control left" href="#myCarousel" data-slide="prev">&lsaquo;</a>
    <a class="carousel-control right" href="#myCarousel" data-slide="next">&rsaquo;</a>
</div>
```

JavaScript Plugins: Popover



JavaScript Plugins: Popover

HTML

```
<a href="#"
      class="btn btn-large btn-danger"
      rel="popover"
      data-content="And here's some amazing content. It's very engaging. right?"
      data-original-title="A Title">
Click to toggle popover
</a>
```

See it in action http://twitter.github.com/bootstrap/javascript.html#popovers

LESS

The dynamic stylesheet language.

What is LESS?

- Maintainable CSS
- Dynamic Behavior
  - Variables
  - Mixins
  - Nested Elements
  - Operations
  - Functions
- Usage
  - Client Side
  - Server Side

LESS: Variables

```
// LESS

@color: #4D926F;

#header {
  color: @color;
}
h2 {
  color: @color;
}
```

```
/* Compiled CSS */

#header {
  color: #4D926F;
}
h2 {
  color: #4D926F;
}
```

LESS: Mixins

```less
// LESS

.rounded-corners (@radius: 5px) {
  border-radius: @radius;
  -webkit-border-radius: @radius;
  -moz-border-radius: @radius;
}

#header {
  .rounded-corners;
}
#footer {
  .rounded-corners(10px);
}
```

```css
/* Compiled CSS */

#header {
  border-radius: 5px;
  -webkit-border-radius: 5px;
  -moz-border-radius: 5px;
}
#footer {
  border-radius: 10px;
  -webkit-border-radius: 10px;
  -moz-border-radius: 10px;
}
```

LESS: Nested

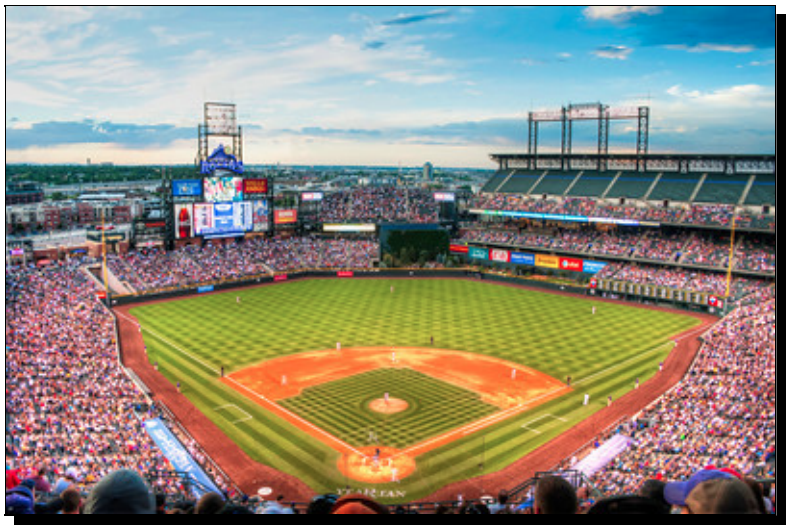
```less
// LESS

#header {
  h1 {
    font-size: 26px;
    font-weight: bold;
  }
  p { font-size: 12px;
    a { text-decoration: none;
      &:hover { border-width: 1px }
    }
  }
}
```

```css
/* Compiled CSS */

#header h1 {
  font-size: 26px;
  font-weight: bold;
}
#header p {
  font-size: 12px;
}
#header p a {
  text-decoration: none;
}
#header p a:hover {
  border-width: 1px;
}
```

Look & Feel



Bootstrap Expo

http://expo.getbootstrap.com

Bootswatch

http://bootswatch.com

WrapBootstrap

https://wrapbootstrap.com

jQuery UI theme

http://addyosmani.github.com/jquery-ui-bootstrap

Font Awesome

http://fortawesome.github.com/Font-Awesome

Bootstrap 3

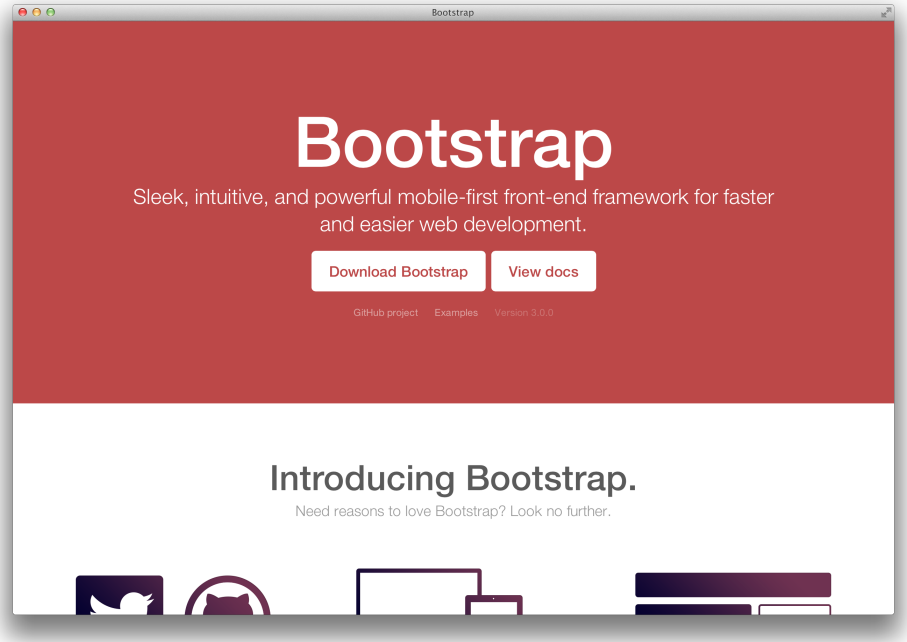
Bootstrap 3 will be mobile-first

Overhaul CSS transitions and reinforce with JavaScript transitions as necessary

Dropping IE7 and FF3.6 support

Combining standard and responsive CSS into a single file

rc.getbootstrap.com →
github.com/twitter/bootstrap/pull/6342



SMACSS

Categorizing CSS rules allows us to see patterns and define best practices around each pattern.

1. Base
2. Layout
3. Module
4. State
5. Theme

Examples of Base Styles

```
html, body, form { margin: 0; padding: 0; }
input[type=text] { border: 1px solid #999; }
a { color: #039; }
a:hover { color: #03C; }
```

SMACSS: Naming Rules

Example classes

```
/* Example Module */
.example { }

/* Callout Module */
.callout { }

/* Callout Module with State */
.callout.is-collapsed { }

/* Form field module */
.field { }

/* Inline layout  */
.l-inline { }
```

http://smacss.com

Jonathan Snook @ HTML5 Denver: June 17, 2013!

UI Refresh Project, 1 Year Later

Mobile Improvements

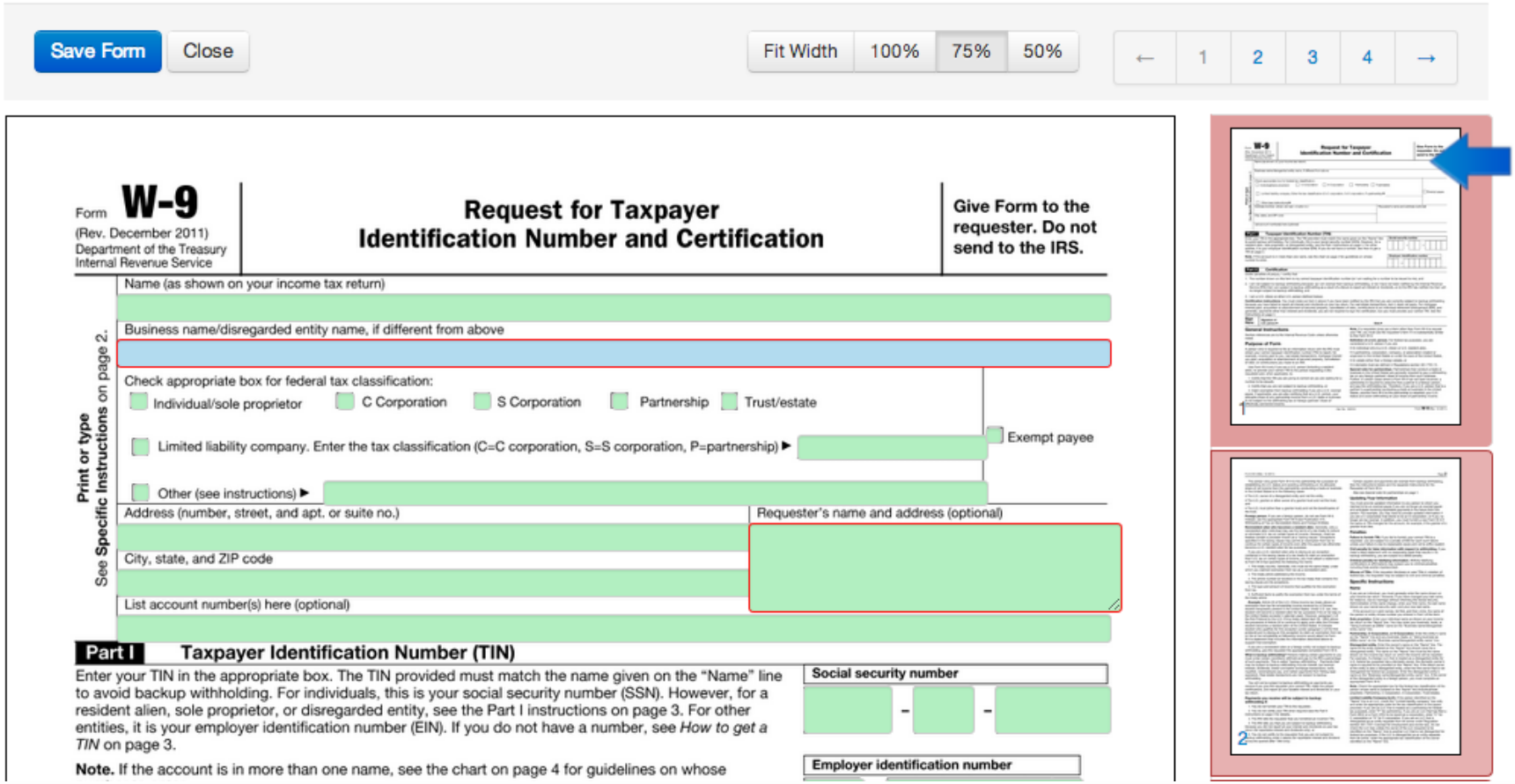
CSS

```
/* Smartphones (portrait and landscape) ----------- */
@media only screen and (min-device-width: 320px) and (max-device-width: 854px) {
    body.mainPage.click-thru {
        padding: 0;
    }
}

/* iPad (portrait and landscape) ----------- */
@media only screen and (min-device-width: 768px) and (max-device-width: 1024px) {
    body.mainPage.click-thru {
        padding-top: 50px;
    }
}
```

UI Refresh Project, 1 Year Later

JavaScript PDF Viewer / Editor



UI Refresh Project, 1 Year Later

My Dashboard

Charts and Widgets, Very Visual

Built With:

- AngularJS
- AngularUI
- AngularUI Bootstrap

PDF Viewer: 200 lines HTML, 2300 lines of JavaScript

My Dashboard: 300 lines HTML, 466 lines of JavaScript

Resources

My Presentations

http://raibledesigns.com/rd/page/publications

Bootstrap

GitHub Project and Google Group

Similar Frameworks

Zurb Foundation and Compass

LESS CSS

http://lesscss.org

Google HTML5 Slides

http://code.google.com/p/io-2012-slides

Dan Vega's Bootstrap 2 Presentation

https://github.com/cfaddict/bootstrap2

<Questions?>

twitter @mraible
www raibledesigns.com
github github.com/mraible
linkedin linkedin.com/in/mraible